Tumor Imaging Positron Emission Tomography (PET) and PET-CT

MCG Health
Ambulatory Care
29th Edition

ACG: A-0098 (AC)
Link to Codes

This MCG care guideline may overlap CMS guidance located in CMS statutes and regulations that are not currently part of MCG's Medicare Compliance content. For Medicare Advantage beneficiaries, before referring to MCG care guidelines, users should first follow all guidance related to this topic located in CMS National Coverage Determinations (NCDs), Local Coverage Determinations (LCDs), and other statutes and regulations. Content used in MCG care guidelines may be used to supplement but does not replace, modify, or supersede existing Medicare regulations or applicable NCDs or LCDs.

Clinical Indications

- Positron emission tomography (PET), with or without simultaneous computed tomography (PET-CT), for tumor imaging may be indicated for **1 or more** of the following(1)(2):
  - Cancer or neoplasm, initial evaluation or staging needed (from diagnosis through initial staging), as indicated by **ALL** of the following(5):
    - Additional imaging information required to assess **1 or more** of the following:
      - Anatomic extent of tumor, if results will assist with selection of optimal antitumor treatment(15)
      - Appropriateness of patient for invasive diagnostic or therapeutic procedure
      - Optimal anatomic location for invasive procedure
    - PET or PET-CT not yet performed (prior to initiation of treatment)
    - Solid tumor malignancy, biopsy-proven or strongly suspected(16)
    - Treatment not yet initiated
    - Type of tumor is **1 or more** of the following:
      - Adrenal cancer(17)(18)(19)(20)(21)(22)(23)
      - Anal cancer(26)(27)(28)
      - Bladder cancer(30)(31)(32)(33)
      - Bone cancer, primary (eg, osteosarcoma or Ewing sarcoma family of tumors)(9)(39)(40)(41)
      - Brain and spinal cord cancer(9)(50)(51)(52)(53)(54)
      - Breast cancer(57)(58)(59)(60)(61)
      - Cervical cancer(70)(71)(72)(73)(74)(75)
      - Colorectal cancer(79)(80)(81)(82)(83)
      - Endometrial cancer(87)(88)(89)
      - Esophageal or gastroesophageal junction cancer(93)(94)(95)(96)(97)(98)
      - Gallbladder cancer and cholangiocarcinoma(102)(103)(104)(105)(106)
      - Gastric cancer(109)(110)
      - Head and neck cancer (nonthyroid, non-central nervous system)(111)(112)(113)(114)(115)(116)(117)
      - Hodgkin lymphoma(9)(127)(128)(129)(130)(131)
      - Kidney cancer(140)(141)(142)(143)
      - Liver cancer(102)(103)(146)(147)(148)

- Lung cancer, non-small cell type(152)(153)(154)(155)(156)
- Lung cancer, small cell type(71)(173)(174)(175)
- Lung nodule, solitary(177)(178)(179)(180)
- Melanoma(184)(185)(186)(187)
- Multiple myeloma, smoldering myeloma, or solitary plasmacytoma(190)(191)(192)
- Neuroblastoma(9)(197)
- Neuroendocrine cancer(23)(201)(202)(203)(204)(205)(206)(207)
- Non-Hodgkin lymphoma(9)(128)(130)(217)(218)(219)(220)(221)(222)(223)(224)
- Ovarian cancer(5)(71)(230)(231)(232)(233)
- Pancreatic cancer(6)(71)(149)(237)(238)(239)
- Paraneoplastic syndrome, including neurologic syndrome(248)(249)
- Pleural mesothelioma, malignant(252)(253)(254)(255)
- Prostate cancer and **1 or more** of the following(257)(258):
  - Equivocal initial bone scan results, and prostate-specific membrane antigen (PSMA), C-11 choline, F-18 fluciclovine, or F-18 sodium fluoride tracer used
  - PSMA tracer (eg, F-18 piflufolastat, F-18 flotufolastat PSMA, Ga-68 PSMA-11) used
- Skin cancer, nonmelanoma (includes basal cell and squamous cell)(265)(266)
- Soft tissue sarcoma, including gastrointestinal stromal tumors(41)(267)(268)(269)(270)(271)
- Thymus cancer(277)(278)
- Thyroid cancer(279)(280)(281)
- Unknown primary cancer(286)(287)(288)(289)(290)(291)
- Vulvar cancer(294)(295)

- Cancer or neoplasm, subsequent evaluation or staging needed (after initiation or completion of treatment), as indicated by **ALL** of the following(296):
  - Need for re-imaging, as indicated by **1 or more** of the following:
    - Imaging required before initiation of tumor-specific therapy (eg, targeted radionuclide therapy)(297)(298)
    - Recurrent or residual disease suspected, as indicated by **1 or more** of the following:
      - Abnormal finding on physical examination
      - Abnormal laboratory test or other imaging study
      - Evaluation of treatment response
      - New symptom
  - Type of tumor is **1 or more** of the following:
    - Adrenal cancer(23)
    - Anal cancer(26)(27)(28)
    - Bladder cancer(30)
    - Bone cancer, primary (eg, osteosarcoma or Ewing sarcoma family of tumors)(39)(40)(41)(296)
    - Brain and spinal cord cancer(50)(51)(52)(53)(54)(55)(301)
    - Breast cancer(57)(59)(296)(304)(305)
    - Cervical cancer(70)(72)(234)(316)(317)(318)
    - Colorectal cancer(79)(80)(81)(82)(296)
    - Endometrial cancer(87)(89)(231)(234)
    - Esophageal or gastroesophageal junction cancer(93)(98)(99)
    - Gastric cancer(109)(328)(329)
    - Gestational trophoblastic neoplasia(332)
    - Head and neck cancer (nonthyroid, non-central nervous system)(111)(114)(116)(117)(296)(333)
    - Hodgkin lymphoma(127)(128)(129)(130)(131)(296)
    - Lung cancer, non-small cell type(152)(154)(344)
    - Lung cancer, small cell type(71)(347)
    - Melanoma(184)(186)(187)(296)(349)(350)
    - Multiple myeloma, smoldering myeloma, or solitary plasmacytoma(5)(190)(191)(192)
    - Neuroendocrine cancer(19)(23)(202)(203)(204)(205)(356)
    - Non-Hodgkin lymphoma(9)(217)(218)(219)(221)(222)(223)(224)(296)
    - Ovarian cancer(71)(230)(233)
    - Pancreatic cancer(237)(238)(239)
    - Pleural mesothelioma, malignant(253)
    - Prostate cancer and **1 or more** of the following(257)(258)(360):
      - Equivocal initial bone scan results, and prostate-specific membrane antigen (PSMA), C-11 choline, F-18 fluciclovine, or F-18 sodium fluoride tracer used
      - PSMA tracer (eg, F-18 piflufolastat, F-18 flotufolastat PSMA, Ga-68 PSMA-11) used
    - Skin cancer, nonmelanoma (includes basal cell and squamous cell)(265)(266)
    - Soft tissue sarcoma, including gastrointestinal stromal tumors(41)(267)(268)(269)(270)(271)
    - Testicular cancer (seminoma)(71)(361)(362)

- Thyroid cancer(279)(280)(363)(364)
- Vulvar cancer(294)(295)

Evidence Summary

Background

PET is a scintigraphic imaging technique providing 3-dimensional information about localized metabolism of labeled trace compounds, most commonly F-18 fluorodeoxyglucose as an analogue for glucose.(3)(4) **(EG 2)** F-18 fluorodeoxyglucose PET is a functional metabolic imaging study indicated for determination of initial (diagnosis and staging) and subsequent (restaging and monitoring) treatment strategies for specific malignancies.(5)(6)(7) **(EG 2)** PET-based metabolic information can be superimposed upon CT-based anatomic data, all obtained from a single scan procedure (PET-CT), resulting in more accurate anatomic localization of functional metabolic information.(6) **(EG 2)** Common pediatric oncologic uses of PET-CT include lymphoma, bone and soft tissue sarcoma, brain tumors, neuroblastoma, and neuroendocrine tumors, among others.(8)(9) **(EG 2)** For pediatric patients, minimization of radiation dose is essential; even so, a dose of 4 mSv may be expected from the PET component, and another 2 mSv from even the lowest-dose CT scan protocols. For each sievert (Sv) of cumulative radiation dose, children up to age 10 years gain an additional 13% to 16% probability of developing a fatal malignancy related to radiation exposure.(8) **(EG 2)**

In a US registry of 22,795 F-18 fluorodeoxyglucose PET studies (84% of which were F-18 fluorodeoxyglucose PET-CT scans) for various malignancies, physicians stated that they changed their intended management in 37% of cases, including avoidance of biopsy in 70% of cases in which biopsy had been anticipated.(10) **(EG 2)** The authors stated that, while PET and PET-CT are being actively used to guide management decisions across a wide spectrum of malignancies, additional studies are needed to validate those decisions, as well as to optimize appropriate sequencing approaches.(10) **(EG 2)** In a subsequent report of 8240 cancer patients who had received 10,497 PET scans (90% of which were PET-CT scans), changes in management occurred in 78% of situations when the scan was judged to predict a worse prognosis vs 40% when the prognosis was improved or unchanged.(11) **(EG 2)**

The radiation dose from a full-body PET-CT scan is significant and may total as high as 32 mSv per study, representing a calculated lifetime additional cancer risk of up to 0.5% in a 20-year-old female and 0.3% in a 20-year-old male.(12) **(EG 2)** Therefore, it may be most appropriate to consider a single PET-CT examination in such situations.(5) **(EG 2)** There is limited evidence regarding the use of PET in children.(13) **(EG 1)**

Criteria

The evidence for the clinical indications found in this guideline includes 252 published peer reviewed articles, 58 specialty society or other evidence-based guidelines, and 3 Cochrane systematic reviews.

For initial evaluation or staging of adrenal cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Specialty society position statements on equivocal adrenal imaging findings state that PET or PET-CT is useful to distinguish potentially malignant lesions when other imaging, such as CT scan or MRI, is inconclusive.(18)(21) **(EG 2)** A systematic review and meta-analysis of 29 studies (2421 patients) evaluating the diagnostic accuracy of PET or PET-CT for differentiating benign from malignant adrenal masses found a pooled sensitivity of 91% and specificity of 91%.(17) **(EG 1)** A systematic review found that PET-CT is useful in characterizing adrenal masses discovered incidentally on CT scan or MRI, as well as in better defining malignant primary and metastatic adrenal lesions.(24) **(EG 1)** A systematic review and meta-analysis of 37 studies evaluating the accuracy of CT, MRI, F-18 fluorodeoxyglucose PET, and PET-CT for the diagnosis of malignancy in incidentally discovered adrenal masses concluded that there was insufficient evidence for the diagnostic value of an individual imaging test to distinguish benign from malignant masses. Additional studies were recommended.(25) **(EG 1)**

For initial evaluation or staging of anal cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert consensus guidelines indicate that PET-CT may be considered during treatment planning.(26)(27)(29) **(EG 2)** A systematic review and meta-analysis of 17 studies found that PET/PET-CT and CT scan had pooled sensitivities of 99% and 67%, respectively, for the initial detection of primary anal cancer. For the detection of inguinal lymph node metastases, PET-CT had a pooled sensitivity of 93% and specificity of 76%. In addition, PET or PET-CT upstaged 5% to 38% of patients and downstaged 8% to 27% of patients; treatment plans were modified in 13% to 59% of patients.(27) **(EG 1)**

For initial evaluation or staging of bladder cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis found that PET-CT had a pooled sensitivity and specificity of 90% and 100%, respectively, for detecting the primary lesion in bladder cancer; fluorodeoxyglucose PET or PET-CT had a pooled sensitivity and specificity of 82% and 89%, respectively, for staging or restaging bladder cancer. Urinary excretion of F-18 fluorodeoxyglucose tracer can obscure PET-CT evaluation of localized bladder malignancy; the authors recommended larger prospective studies to further evaluate the role of fluorodeoxyglucose PET or PET-CT in the detection of bladder cancer.(34) **(EG 1)** A systematic review and meta-analysis of 14 studies (785 patients) of newly diagnosed bladder cancer evaluated the diagnostic accuracy of PET-CT for lymph node metastases and found a pooled sensitivity and specificity of 57% and 92%, respectively. Additional studies were recommended to confirm the findings.(35) **(EG 1)** An expert consensus guideline indicates that PET-CT may be useful for initial staging in patients with muscle-invasive bladder cancer,(30) although specialty society guidelines do not specify any role for PET or PET-CT in diagnosis or management.(36)(37) **(EG 2)** The use of F-18 fluorodeoxyglucose PET-CT may be helpful in patients with bladder cancer at high risk for metastatic disease (eg, T3b disease) in order to potentially avoid radical surgical procedures.(38) **(EG 2)**

For initial evaluation or staging of bone cancer, primary (eg, osteosarcoma or Ewing sarcoma family of tumors), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Expert consensus guidelines, a meta-analysis, and review articles indicate that PET is an option for primary disease staging.(39)(42)(43)(44)(45)(46)(47) **(EG 1)** A meta-analysis of 23 studies (524 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET and PET-CT in Ewing sarcoma family of tumors reported that pooled sensitivities and pooled specificities varied by indication; for diagnosis (13 trials), the pooled sensitivity was 86% and the pooled specificity was 80%.(48) **(EG 1)** A systematic review and meta-analysis of 26 studies (798 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET or PET-CT for diagnosing, staging, and recurrence monitoring of osteosarcoma found that the imaging method had a sensitivity of 100% for detecting primary lesions; a sensitivity and specificity of 90% and 96%, respectively, for detecting metastases; and a sensitivity and specificity of 91% and 93%, respectively, for identifying recurrent disease.(49) **(EG 1)**

For initial evaluation or staging of brain and spinal cord cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review including 24 meta-analyses assessing the efficacy of PET-CT for brain cancer evaluation reported a range of pooled sensitivities (38% to 95%) and specificities (74% to 88%) depending upon the tracer used. The authors concluded that the efficacy of PET-CT varied based on tracer and advised robust prospective trials for further evaluation.(55) **(EG 1)** A review article and evidence map of clinical studies using PET and PET-CT in glioma found mostly studies assessing diagnostic or prognostic performance, but little evidence demonstrating impact upon diagnostic thinking, therapeutic decisions, or clinical outcomes relevant to patients.(56) **(EG 2)** Malignant glioma, lymphoma, meningioma, or brain metastases may show significantly elevated fluorodeoxyglucose uptake, and in such cases, the most active area on PET-CT is often the most suitable target for biopsy.(50)(51) **(EG 2)** In addition, fluorodeoxyglucose uptake correlates with tumor grade and prognosis, and may help distinguish radiation necrosis and pseudo-progression from active malignancy.(50)(52) **(EG 2)**

For initial evaluation or staging of breast cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A meta-analysis analyzed the use of F-18 fluorodeoxyglucose PET to diagnose primary tumors in women suspected of having breast cancer and found a range of sensitivities from 48% to 96%; F-18 fluorodeoxyglucose PET seemed to show diminished sensitivity for tumors less than 10 mm in diameter.(62) **(EG 1)** PET is insensitive to detection of micrometastases in the axillary lymph nodes, when compared with surgical biopsy and histologic assessment, and should not be used for staging in that area.(63)(64) **(EG 1)** A systematic review and meta-analysis of 21 studies (1095 patients) evaluating the efficacy of MRI or PET-CT for the diagnosis of axillary lymph node status in breast cancer patients found pooled sensitivities of 82% and 64%, respectively, and specificities of 93% and 93%, respectively.(58) **(EG 1)** A meta-analysis of 6 studies including 609 patients compared PET or PET-CT with conventional imaging, including chest x-ray, abdominal ultrasound, and bone scan, and found that PET or PET-CT had significantly higher sensitivity (99% vs 57%, respectively) and specificity (95% vs 88%, respectively) in detecting distant metastases.(65) **(EG 1)** A systematic review and meta-analysis of 7 studies including 171 patients with estrogen receptor-positive breast cancer examining the diagnostic performance of F-18 fluoroestradiol PET-CT as compared with F-18 fluorodeoxyglucose PET-CT for detecting metastases found no significant difference in sensitivity between the 2 PET tracers in a patient-based analysis (97% for F-18 fluorodeoxyglucose vs 94% for F-18 fluoroestradiol); however, F-18 fluoroestradiol was more sensitive than F-18 fluorodeoxyglucose in a lesion-based analysis, particularly when restaging disease (98% for F-18 fluoroestradiol vs 81% for F-18 fluorodeoxyglucose). The authors noted that the analysis was limited by the retrospective study design, heterogeneity, and small number of patients in the included studies; larger prospective studies were recommended.(66) **(EG 1)** An expert consensus guideline states that fluorodeoxyglucose PET may be appropriate when standard staging imaging is indeterminate and may also be used alongside standard imaging to detect regional lymph nodes and metastatic disease.(57) **(EG 2)** Review articles state that PET or PET-CT is most useful when standard staging measures for breast cancer are inconclusive, especially in the setting of locally advanced, inflammatory, or metastatic disease.(59)(67)(68)(69) **(EG 2)** However, due to physiologic fluorodeoxyglucose uptake by brain tissue, a principal limitation of PET-CT is lack of sensitivity for metastatic disease to this area.(59) **(EG 2)**

For initial evaluation or staging of cervical cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review studies and expert opinion indicate that PET should be a standard part of the evaluation of patients with stage IB1 or higher due to its ability to identify lymphatic and distant metastases in primary and recurrent disease.(70)(71)(72)(76) **(EG 2)** A systematic review and meta-analysis of 67 studies compared different imaging modalities for detection of metastatic lymph nodes in patients with cervical cancer and found that PET or PET-CT had sensitivity of 66% and specificity of 97%, that CT scan had sensitivity of 57% and specificity of 91%, and that MRI had sensitivity of 54% and specificity of 93%.(77) **(EG 1)** The ability of PET-CT to better evaluate locoregional disease and distant nodal metastases has resulted in therapy management changes, including radiation therapy planning, although incremental effects of such testing on longer-term outcomes are unknown.(78) **(EG 2)**

For initial evaluation or staging of colorectal cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert consensus guidelines and a systematic review do not recommend the routine use of PET or PET-CT for staging.(80)(84) **(EG 1)** If CT scan or MRI findings are inconclusive for metastatic disease, PET or PET-CT may be helpful.(79)(80)(84)(85) **(EG 2)** In addition, if prior imaging suggests potentially surgically curable M1 disease, PET or PET-CT may be helpful in ruling out unexpected metastases that would preclude such surgery.(79)(80)(81)(86) **(EG 2)**

For initial evaluation or staging of endometrial cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of 19 studies (1431 patients) evaluating the accuracy of PET or PET-CT in detecting nodal metastases in patients with endometrial cancer found a pooled sensitivity and specificity of 68% and 96%, respectively. The authors identified heterogeneity in study samples and methodologies as limitations.(90) **(EG 1)** A multicenter prospective observational study of 203 patients with high-risk endometrial cancer reported that PET-CT read by a centralized reader found that 12% of patients had distant metastatic disease and that PET-CT demonstrated a sensitivity, specificity, positive predictive value, and negative predictive value of 65%, 99%, 86%, and 95%, respectively. The authors concluded that PET-CT may offer an accurate noninvasive approach to evaluating distant metastatic disease.(91) **(EG 2)** An observational study of 362 patients with endometrial cancer who were lymph node negative on preoperative MRI found that PET-CT had sensitivity of 18.5% and specificity of 94% for detection of lymph node metastasis. The authors concluded that preoperative PET-CT had a low value in predicting lymph node metastases in patients with endometrial cancer who were node negative on MRI, and additional studies were recommended.(92) **(EG 2)** An expert consensus guideline states that PET-CT is an option for initial evaluation of endometrial cancer with suspected metastatic disease.(89) **(EG 2)**

For initial evaluation or staging of esophageal or gastroesophageal junction cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** PET, as a complement to CT scan and endoscopic ultrasound, increases the detection of metastasis and helps avoid futile surgical treatment.(97)(99)(100) **(EG 2)** PET-CT is preferred over PET alone for staging.(93)(95)(99) **(EG 1)** A prospective study of 491 patients with potentially resectable (M0) esophageal cancer based on conventional imaging (CT scan plus endoscopic ultrasound) found that PET-CT changed management in 24% of patients; 21% of patients were classified as a higher stage after PET-CT was performed.(101) **(EG 2)**

For initial evaluation or staging of gallbladder cancer or cholangiocarcinoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** For gallbladder cancer, expert consensus guidelines state that the routine use of PET-CT in the preoperative setting has not been established in prospective trials. PET-CT may be considered when there are equivocal findings on CT or MRI.(102)(105) **(EG 2)** A systematic review and meta-analysis of 21 studies (495 patients) evaluating the diagnostic accuracy of F-18 fluorodeoxyglucose PET and PET-CT for the evaluation of gallbladder cancer found a pooled sensitivity and specificity of 87% and 78%, respectively.(107) **(EG 1)** For extrahepatic cholangiocarcinoma, an expert consensus guideline states that PET-CT may be considered when there are equivocal findings on CT or MRI, or for patients being evaluated for resection to evaluate the presence of distant extrahepatic disease.(102) **(EG 2)** A meta-analysis of 23 studies found that F-18 fluorodeoxyglucose PET and PET-CT had pooled sensitivity and specificity of 81% and 89%, respectively, in detecting cholangiocarcinoma; sensitivity was higher for intrahepatic cholangiocarcinoma as compared with extrahepatic cholangiocarcinoma.(108) **(EG 1)**

For initial evaluation or staging of gastric cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** An expert consensus guideline states that PET scan is generally not utilized for preoperative staging because it has a lower accuracy rate than CT scan. In addition, PET has a significantly lower sensitivity, although increased specificity, than CT scan for the detection of lymph node involvement. However, PET-CT has a significantly increased accuracy rate (68%) for preoperative staging as compared with PET scan (47%) or CT scan (53%) alone.(109) **(EG 2)** A specialty society clinical practice guideline does not routinely recommend PET-CT for initial staging of gastric cancer, noting that, while PET-CT may improve initial staging by identifying involved lymph nodes or occult metastatic disease in some patients, PET-CT may not be as useful for patients with mucinous or diffuse tumors due to lower tracer uptake.(110) **(EG 2)**

For initial evaluation or staging of head and neck cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** An expert consensus guideline supports the use of PET-CT for evaluation of selected clinical stage III or IV head and neck cancers (eg, glottis, oral cavity, pharynx, sinus) because it may alter management by upstaging patients.(114) **(EG 2)** Clinical studies and reviews indicate that fluorodeoxyglucose PET is effective for initial staging in adult(112)(113)(118)(119)(120) and pediatric patients with head and neck cancer.(121)(122) **(EG 2)** A meta-analysis of 9 studies indicated that pooled sensitivity and specificity of PET-CT for diagnosis of head and neck cancer were 89.3% and 89.5%, respectively, while diagnosis with MRI and CT scan had sensitivity and specificity of only 71.6% and 78.0%, respectively.(123) **(EG 1)** A meta-analysis of 20 studies with a total of 2396 patients found that F-18 fluorodeoxyglucose PET and PET-CT had pooled sensitivity and specificity of 89% and 96%, respectively, for the detection of lymph node metastasis and distant metastasis (N and M staging) in nasopharyngeal cancer.(124) **(EG 1)** Review articles concluded that fluorodeoxyglucose PET is more effective than MRI or CT for nodal and metastatic staging in nasopharyngeal cancer.(125)(126) **(EG 2)**

For initial evaluation or staging of Hodgkin lymphoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** For Hodgkin lymphoma, practice guidelines recommend PET or PET-CT for initial staging.(127)(128) **(EG 2)** Specialty society guidelines recommend F-18 fluorodeoxyglucose PET-CT instead of CT scan for the initial staging of fluorodeoxyglucose-avid Hodgkin lymphoma.(128)(132) **(EG 2)** Systematic reviews and expert opinion indicate that PET is effective in routine initial staging evaluation of Hodgkin lymphoma,(133) including estimation of bone marrow involvement.(134)(135)(136)(137)(138) **(EG 2)** Studies have found that the sensitivity and specificity of F-18 fluorodeoxyglucose PET are significantly higher than those of contrast CT scan alone, with resulting improvements in staging; however, improved survival has not been documented.(139) **(EG 2)**

For initial evaluation or staging of kidney cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Some but not all authors have demonstrated that PET seems to be effective in detection of distant metastases; most agree that it is likely to be helpful as a complementary imaging modality in equivocal cases.(140)(141)(142) **(EG 2)** However, specialty society guidelines indicate that PET has not yet been widely accepted as a standard imaging tool for either diagnosis of kidney cancer or surveillance of relapse after nephrectomy.(141)(143)(144) **(EG 2)** A meta-analysis of 14 studies indicated that pooled sensitivity and specificity of PET for renal cancer lesions were 62% and 88%, respectively, while use of PET-CT to detect extrarenal lesions had pooled sensitivity and specificity of 91% and 88%, respectively, with good consistency across multiple studies.(145) **(EG 1)**

For initial evaluation or staging of liver cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis of 9 studies found that F-18 fluorodeoxyglucose PET and PET-CT had pooled sensitivity of 95% for the detection of intrahepatic cholangiocarcinoma.(108) **(EG 1)** Sensitivity of PET is estimated to be only 55% to 64% for detection of primary hepatocellular tumors.(103)(149) **(EG 2)** For hepatocellular carcinoma, expert consensus guidelines and reviews indicate that PET is not adequate for full tumor delineation(150) and recommend a multiphasic protocol of CT scan or MRI to fully characterize perfusion characteristics, extent and number of lesions, vascular anatomy, and presence of extrahepatic disease.(102) **(EG 2)** For metastatic lesions, a review indicates that PET-CT is an effective imaging method for detecting hepatic metastases from a variety of primary malignancies,(146) although it is unclear whether it is superior to contrast-enhanced CT or MRI.(151) **(EG 2)**

For initial evaluation or staging of lung cancer (non-small cell type), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Specialty society guidelines and systematic reviews indicate that PET has a well-established role in diagnosis and staging of non-small cell lung cancer.(152)(153)(154)(157)(158)(159)(160)(161)(162) **(EG 2)** For assessment of the mediastinum, however, a meta-analysis showed that sensitivity is only 72%,(163) and positive mediastinal findings require pathologic confirmation.(164)(165)(166) **(EG 2)** A specialty society guideline states that PET-CT is sufficient for mediastinal staging when tumors are smaller than 3 cm, are peripherally located, and have no lymph node involvement.(167) **(EG 2)** In a randomized controlled trial of patients with diagnosed non-small cell lung cancer, preoperative staging with PET-CT identified more patients with mediastinal and extrathoracic disease than did conventional staging with abdominal CT scan and bone scan, sparing more patients from stage-inappropriate surgery. However, some patients had false-positive mediastinal nodes on PET-CT, which could inadvertently exclude such patients from curative surgery unless there was biopsy confirmation.(168) **(EG 1)** Another randomized controlled trial on the use of PET-CT for preoperative staging in non-small cell lung cancer found that, while there was no effect on overall mortality, patients receiving PET-CT had fewer total and unnecessary thoracotomies.(169) **(EG 1)** PET-CT is also a cornerstone for planning radiation therapy in such patients.(170) **(EG 2)** Two meta-analyses, of 7 and 17 studies assessing the relative imaging value of PET-CT and bone scan for finding bony metastases in patients with mostly non-small cell lung cancer, found pooled sensitivities of 93% and 92% for PET-CT and 87% and 86% for bone scan. Pooled lesion-based sensitivity and specificity from the smaller meta-analysis were 93% and 91%, respectively, for PET-CT and 92% and 57%, respectively, for bone scan.(171)(172) **(EG 1)**

For initial evaluation or staging of lung cancer (small cell type), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Expert consensus guidelines indicate that PET-CT improves accuracy of initial staging, especially in evaluation of the mediastinum.(157)(159)(174) **(EG 2)** A systematic review suggests that sensitivity of PET-CT staging approaches 100%, with specificity of over 90%, and PET-CT alters management in at least 28% of patients; however, incremental effects of such testing and management changes on longer-term outcomes are unknown.(176) **(EG 1)** Pathologic confirmation is recommended for any stage-altering lesions detected by PET-CT.(174) **(EG 2)** PET-CT is also helpful in radiotherapy planning.(174) **(EG 2)**

For initial evaluation or staging of lung nodule (solitary), evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Reviews of clinical studies and expert opinion indicate that PET is usually appropriate for characterizing indeterminate pulmonary nodules 6 mm or larger identified on chest CT scan.(177)(178) **(EG 2)** A specialty society recommends PET-CT to evaluate a pulmonary nodule in a patient with a greater than or equal to 10% risk of malignancy using the Brock model if the nodule is larger in size than the local PET-CT threshold.(181) **(EG 2)** Significantly high lesion uptake values on PET are associated with a malignancy probability of 90%, while negative PET lesions have less than a 5% probability of malignancy.(182) **(EG 2)** A meta-analysis of 70 studies found that F-18 fluorodeoxyglucose PET-CT has less specificity in geographic areas with endemic infectious lung disease.(183) **(EG 1)** Meta-analyses of studies evaluating the efficacy of PET-CT for the diagnosis of pulmonary nodules reported sensitivity and specificity ranges of 80% to 82% and 75% to 81%, respectively, for differentiating malignant from benign pulmonary nodules.(179)(180) **(EG 1)**

For initial evaluation or staging of melanoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Expert consensus guidelines and a review article recommend that PET-CT scan is most useful as a whole-body imaging tool for initial evaluation of stage III or IV disease.(184)(188) **(EG 2)** PET is generally not utilized for the detection of subclinical metastatic disease in patients with clinically localized melanoma because the yield is low, with poor sensitivity.(184) **(EG 2)** For staging of distant metastases in patients with melanoma, a meta-analysis found that PET-CT had the highest sensitivity and specificity when compared with ultrasonography, CT scan, or PET alone.(189) **(EG 1)**

For initial evaluation or staging of multiple myeloma, smoldering myeloma, or solitary plasmacytoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Expert consensus guidelines and a meta-analysis indicate that PET-CT is an option during initial diagnostic evaluation, especially for assessment of possible extramedullary involvement,(46)(193)(194)(195) and for unexplained signs or symptoms.(190)(196) **(EG 1)**

For initial evaluation or staging of neuroblastoma, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of 40 studies (1134 patients) found that, for the diagnosis of neuroblastoma, metaiodobenzylguanidine scintigraphy had sensitivity of 79% and specificity of 84%, and PET-CT had sensitivity of 89% and specificity of 71%.(197) **(EG 1)** While metaiodobenzylguanidine scintigraphy is the preferred imaging study for staging of neuroblastoma, F-18 fluorodeoxyglucose PET-CT may be useful for tumors that are negative on scintigraphy or have lower than expected tumor burden based on clinical symptoms.(198)(199)(200) **(EG 2)**

For initial evaluation or staging of neuroendocrine cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis of 22 studies with a total of 2105 patients found that PET-CT (using somatostatin receptor radiotracers) had pooled sensitivity of 93% and pooled specificity of 96% for detection of neuroendocrine tumors in the thorax and abdomen.(208) **(EG 1)** A systematic review and meta-analysis of 14 studies (1561 patients) found that use of somatostatin PET-CT resulted in a change of treatment plans for 44% of patients with neuroendocrine tumors.(209) **(EG 1)** A systematic review of 10 studies with a total of 329 patients found F-18 fluorodeoxyglucose PET and PET-CT to be useful in detecting nodal and metastatic disease in Merkel cell carcinoma.(210) **(EG 1)** A systematic review of 9 studies comparing the efficacy of 68Ga-DOTATATE PET-CT and conventional imaging (eg, CT, MRI) found that 68Ga-DOTATATE PET-CT had estimated sensitivity and specificity of 91% and 91%, respectively, for the detection of pulmonary and gastropancreatic neuroendocrine tumors.(211) **(EG 1)** A meta-analysis of 17 studies (629 patients) found that F-18 fluorodeoxyglucose PET-CT had a sensitivity and specificity of 85% and 55%, respectively, and somatostatin receptor analogue PET-CT had a sensitivity and specificity of 95% and 87%, respectively, for the diagnosis of metastatic pheochromocytoma and paraganglioma.(212) **(EG 1)** A systematic review and meta-analysis identified 6 studies that evaluated 68Ga-DOTATATE, 68Ga-DOTATOC, or 68Ga-DOTANOC PET or PET-CT during the initial diagnosis of neuroendocrine tumors and reported a pooled sensitivity of 91% and a pooled specificity of 94%. Of 14 studies that evaluated use for staging and restaging, PET or PET-CT had a sensitivity range of 78% to 100% and a specificity range of 50% to 100% for the detection of primary and/or metastatic lesions.(213) **(EG 1)** An observational study of 43 patients found that GLP-1R PET-CT, CT, MRI, endoscopic ultrasound, and somatostatin receptor scintigraphy had lesion-based sensitivities of 98%, 73%, 56%, 85%, and 16%, respectively, for the detection of insulinomas.(214) **(EG 2)** An observational study of 30 patients found that 68Ga-DOTATATE PET-CT, 18F-FDOPA PET-CT, and conventional imaging had lesion-based sensitivities of 93%, 89%, and 76%, respectively, for the detection of pheochromocytomas and paragangliomas.(215) **(EG 2)** Review articles indicate that use of alternative positron-emitting radiotracers, such as 18F-FDOPA for documented carcinoid tumors or 68Ga-DOTANOC for gastroenteropancreatic neuroendocrine tumors, during PET is highly effective in staging disease.(19)(202)(216) **(EG 2)** Specialty society and expert consensus guidelines indicate that 68Ga-DOTATATE PET is preferred over indium-111 somatostatin receptor scintigraphy for the initial evaluation or staging of gastroenteropancreatic neuroendocrine tumors (eg, carcinoid, gastrinoma, glucagonoma, insulinoma, neuroendocrine tumor of unknown primary site, VIPoma).(23)(205) **(EG 2)** A specialty society guideline recommends F-18 fluorodeoxyglucose PET-CT as the preferred imaging study in patients with known metastatic pheochromocytoma or paraganglioma.(204) **(EG 2)**

For initial evaluation or staging of non-Hodgkin lymphoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** For B-cell and T-cell lymphoma, practice guidelines recommend PET or PET-CT for initial staging.(128)(132)(217)(219)(225)(226) **(EG 2)** For chronic lymphocytic leukemia, PET is generally not useful unless directing lymph node biopsy when there is a suspicion of Richter transformation.(220) **(EG 2)** For primary cutaneous B-cell lymphoma, an expert consensus guideline notes that PET-CT may be appropriate for initial staging; however, a high rate of false positives for occult systemic disease may lead to diagnostic confusion.(218) **(EG 2)** Systematic reviews and expert opinion indicate that fluorodeoxyglucose PET is effective in routine initial staging evaluation of non-Hodgkin lymphoma,(133) including estimation of bone marrow involvement and aggressive cell types (especially diffuse large B-cell lymphoma and follicular lymphoma).(134)(135)(137)(138)(227) **(EG 2)** Studies have found that the sensitivity and specificity of F-18 fluorodeoxyglucose PET are significantly higher than those of contrast CT scan alone, with resulting improvements in staging; however, improved survival has not been documented.(139) **(EG 2)** For follicular lymphoma, a systematic review of 7 staging studies (349 patients) found that, although F-18 fluorodeoxyglucose PET appears to be more sensitive than CT with regard to upstaging, the studies reviewed had numerous methodological errors. The authors noted that F-18 fluorodeoxyglucose PET has an unacceptably low sensitivity for detecting bone marrow involvement and that increased tracer uptake is absent in the majority of histologically proven follicular lymphoma of the gastrointestinal tract. Additional studies were recommended.(228)(229) **(EG 1)**

For initial evaluation or staging of ovarian cancer (including epithelial ovarian cancer, fallopian tube cancer, and primary peritoneal cancer), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** PET is not sensitive enough to identify microscopic or small-volume (ie, less than 1 cm^3) intra-abdominal spread; consequently, its use in initial staging is limited,(71) although some review articles maintain that it is useful for initial staging.(5)(234)(235)(236) **(EG 2)** A specialty society guideline states that PET may be utilized in selected cases of germ cell tumors.(233) **(EG 2)** An expert consensus guideline indicates that PET-CT is probably most useful during initial staging to evaluate indeterminate findings, especially when results may alter management.(230) **(EG 2)** A meta-analysis concluded that PET-CT was more accurate than CT scan or MRI alone for detection of lymph node metastases in patients with ovarian cancer.(232) **(EG 1)**

For initial evaluation or staging of pancreatic cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of 52 studies (5399 patients) compared various imaging modalities for the diagnosis of pancreatic ductal adenocarcinoma and found that PET had sensitivity of 89% and specificity of 70%, CT scan had sensitivity of 90% and specificity of 87%, MRI had sensitivity of 93% and specificity of 89%, endoscopic ultrasound had sensitivity of 91% and specificity of 86%, and transabdominal ultrasound had sensitivity of 88% and specificity of 94%.(240) **(EG 1)** A meta-analysis of 39 studies found that PET was less effective at nodal staging and detecting liver metastasis, with pooled sensitivities of 64% and 67%, respectively, and pooled specificities of 81% and 96%, respectively.(241) **(EG 1)** PET-CT provides additional diagnostic benefit as an adjunct to contrast-enhanced CT scan alone, primarily during initial staging and treatment planning.(150)(242)(243)(244) **(EG 2)** In a study of 82 patients with potentially resectable pancreatic cancer, PET-CT results changed management in 11% of cases.(245) **(EG 2)** However, evidence for effectiveness during subsequent treatment such as restaging or monitoring for recurrence is limited.(149)(150)(242) **(EG 2)** A specialty society practice guideline states that PET-CT does not add much staging information in most patients with resectable disease(246); however, it may be helpful in detecting previously unknown metastatic disease, thus preventing unnecessary surgery.(6) **(EG 2)** A practice guideline supports the use of PET-CT in patients with suspected pancreatic cancer with or without obstructive jaundice when CT scan is nondiagnostic.(247) **(EG 2)**

For initial evaluation or staging of paraneoplastic syndrome, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A clinical study and specialty society guidelines indicate that PET-CT is effective in a sizable minority of patients for finding malignancy in those with suspected paraneoplastic syndrome who have undergone a negative standard evaluation.(248)(249)(250) **(EG 2)** A systematic review and meta-analysis of 21 studies (1293 patients) evaluating the diagnostic accuracy of PET or PET-CT for underlying malignancy in patients with paraneoplastic syndrome found a pooled sensitivity and specificity of 81% and 88%, respectively. Subgroup analysis of 528 patients with exclusive neurologic paraneoplastic symptoms found a pooled sensitivity and specificity of 89% and 83%, respectively.(251) **(EG 1)**

For initial evaluation or staging of pleural mesothelioma (malignant), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Specialty society practice guidelines support the use of PET-CT for the initial staging of malignant pleural mesothelioma. PET-CT is not recommended after talc pleurodesis because of false-positive results.(252)(254) **(EG 2)** An expert consensus guideline recommends PET-CT as part of the surgical evaluation of patients with clinical stage I to clinical stage IIIA malignant pleural mesothelioma and epithelioid histology.(253) **(EG 2)** A national specialty society guideline states that, for both stage II and III malignant pleural mesothelioma, PET-CT has sensitivity and specificity of 100% and 100%, respectively, as compared with MRI, which has sensitivity and specificity of 88% and 88%, respectively, for stage II, and sensitivity and specificity of 91% and 100%, respectively, for stage III disease.(256) **(EG 2)**

For initial evaluation or staging of prostate cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of C-11 choline and F-18 fluorocholine PET and PET-CT in prostate cancer staging and restaging reported a pooled sensitivity of 85% and a pooled specificity of 88%. Most of the studies identified were retrospective, and concerns regarding the reference standards employed were noted; the authors concluded that the evidence for clinical validity and clinical utility remains to be developed.(259) **(EG 1)** A systematic review and meta-analysis evaluating the diagnostic accuracy of prostate-specific membrane antigen (PSMA) PET-CT found that, for localized prostate cancer, the pooled sensitivity and specificity were 71% and 92%, respectively, and for lymph node-metastatic disease, the sensitivity and specificity were 57% and 96%, respectively. The analyses were limited by high levels of heterogeneity between studies.(260) **(EG 2)** A randomized trial including 277 individuals with intermediate-risk to high-risk prostate cancer who underwent radical prostatectomy and lymph node dissection found that Ga-68 PSMA-11 PET had a sensitivity of 0.4 and a specificity of 0.95 based on surgical pathology report.(261) **(EG 1)** Most prostate cancers are well differentiated and grow slowly with low PET positivity, but the minority of patients with aggressive, rapidly growing tumors may demonstrate greater PET positivity.(262) **(EG 2)** A review article states that, although choline PET-CT has a low sensitivity and specificity for differentiating prostate cancer from normal prostate tissue or hyperplasia, it has a high specificity for detecting bone metastases.(263) **(EG 2)** A specialty society guideline based on a systematic review states that PET or PET-CT may be offered to patients with high-risk prostate cancer and negative or equivocal initial imaging (ie, CT, bone scan, or MRI).(264) **(EG 2)** A specialty society states that PSMA PET imaging may be used in the initial evaluation of bone and soft tissue metastatic disease, to assess PSA recurrence, or to evaluate disease progression due to its increased sensitivity and specificity as compared with conventional cross-sectional imaging; C-11 choline and F-18 fluciclovine may be used to evaluate small-volume recurrent disease in the bone and soft tissue, while F-18 sodium fluoride is specific to bone metastases and its use is limited to when bone scan results are equivocal.(257) **(EG 2)**

For initial evaluation or staging of skin cancer (nonmelanoma), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A review of 40,863 PET studies at 1386 centers, of which 1057 PET studies were for nonmelanoma skin cancer, indicated that PET findings resulted in management changes for 31% of such patients, but incremental effects of such testing and management changes on longer-term outcomes are unknown.(265) **(EG 2)** For nonmelanoma skin cancer patients undergoing PET for initial staging, PET results changed management in 38%, but incremental effects of such testing and management changes on longer-term outcomes are unknown.(265) **(EG 2)**

For initial evaluation or staging of soft tissue sarcoma, including gastrointestinal stromal tumors, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Fluorodeoxyglucose PET-CT may be useful for initial staging of soft tissue sarcoma, especially if CT scan or MRI is indeterminate.(42)(267)(269)(272) **(EG 2)** A systematic review and meta-analysis of 14 studies (755 patients) comparing the diagnostic efficacy of F-18 fluorodeoxyglucose PET and PET-CT found sensitivities of 96% and 91%, respectively, and specificities of 71% and 85%, respectively, for the diagnosis of musculoskeletal soft tissue malignancies.(273) **(EG 1)** For gastrointestinal stromal tumors, PET may be helpful in differentiating active disease from necrotic or scar tissue and malignant from benign tissue; however, it is not to be considered a substitute for CT scan.(268) **(EG 2)** PET-CT is also helpful in early assessment of metastatic disease.(274)(275)(276) **(EG 2)**

For initial evaluation or staging of thymus cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Preliminary studies suggest that PET-CT may be effective at differentiating thymic carcinoma from other entities.(277) **(EG 2)** In addition, PET-CT enables more accurate preoperative staging and better guidance of resection for resectable thymic epithelial tumors.(278) **(EG 2)** However, further study on the role of PET-CT in restaging must be undertaken.(278) **(EG 2)**

For initial evaluation or staging of thyroid cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles of clinical studies indicate that PET is effective in initial evaluation of disease extent in anaplastic cancer and oncocytic carcinoma (Hurthle cell carcinoma),(282) as well as in medullary, papillary, and follicular subtypes with elevated thyroglobulin but negative whole-body radioiodine scan.(282)(283) **(EG 2)** In the setting of differentiated thyroid cancer, a study of 327 patients found that sensitivity and specificity of PET-CT were 92% and 95%, respectively, and such imaging resulted in management changes in 43% of patients and a change in surgical approach in 20%; however, incremental effects of such testing and management changes on longer-term outcomes are unknown.(284) **(EG 2)** A specialty society statement does not recommend the routine use of PET or PET-CT in the preoperative setting for thyroid cancer.(285) **(EG 2)** An expert consensus guideline notes that PET-CT may be appropriate for the initial evaluation of patients diagnosed with anaplastic thyroid carcinoma or medullary thyroid carcinoma on fine needle biopsy.(279) **(EG 2)**

For initial evaluation or staging of cancer of unknown primary site, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of 12 studies (383 patients) evaluating the diagnostic accuracy of somatostatin receptor PET-CT in patients with metastatic neuroendocrine tumors and unknown primary found that detection rates ranged from 29% to 83%, with a pooled estimate of 56%. The authors advised future robust trials to confirm the results.(292) **(EG 1)** PET can detect the primary cancer site in up to 20% to 40% of cases,(286) although PET-CT probably provides higher diagnostic accuracy than either method alone.(287)(288) **(EG 1)** PET-CT may be indicated for the evaluation and staging of patients who have head and neck squamous cell carcinoma with an unknown primary site.(293) **(EG 2)** An expert consensus guideline states that, although PET and PET-CT have been shown to be useful for diagnosis and staging, larger studies with longer follow-up are recommended to determine their clinical utility in patients with occult primary tumors. However, PET-CT may be warranted in some situations, especially when considering local or regional therapy.(289) **(EG 2)** It is important that the diagnosis of malignancy be established prior to recommended use of PET-CT.(288) **(EG 1)**

For initial evaluation or staging of vulvar cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An expert society guideline states that PET-CT is usually appropriate in the pretreatment staging of vulvar cancers larger than 4 cm or any size tumor with more than minimal involvement of the urethra, vagina, or anus.(295) **(EG 2)** An expert consensus guideline notes that PET-CT is appropriate for the initial evaluation of T2 or larger cancers or if metastasis is suspected.(294) **(EG 2)**

For subsequent evaluation or staging of adrenal cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Specialty society guidelines recommend F-18 fluorodeoxyglucose PET-CT in the post-treatment follow-up of patients with locoregional unresectable or metastatic disease.(23)(299) **(EG 2)**

For subsequent evaluation or staging of anal cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Expert consensus guidelines indicate that PET or PET-CT may be used for restaging or assessing evidence of progression after primary therapy.(26)(27)(29) **(EG 2)** A case series found that PET-CT, as compared with PET or CT alone, was incrementally helpful in detecting or excluding persistent or recurrent disease after initial therapy, but incremental effects of such testing on longer-term outcomes are unknown.(300) **(EG 2)**

For subsequent evaluation or staging of bladder cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis found that fluorodeoxyglucose PET or PET-CT had pooled sensitivity and specificity of 82% and 89%, respectively, for staging or restaging bladder cancer.(34) **(EG 1)** An expert consensus guideline indicates that PET-CT is an option for follow-up of patients with bladder cancer when concerns for metastatic disease exist(30); however, other specialty society guidelines currently do not specify any role for PET or PET-CT in diagnosis or management.(32)(36)(37) **(EG 2)**

For subsequent evaluation or staging of bone cancer, primary (eg, osteosarcoma or Ewing sarcoma family of tumors), evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of 26 studies (798 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET or PET-CT for diagnosis, staging, and recurrence monitoring of osteosarcoma found that the imaging method had a sensitivity of 100% for detecting primary lesions; a sensitivity and specificity of 90% and 96%, respectively, for detecting metastases; and a sensitivity and specificity of 91% and 93%, respectively, for identifying recurrent disease.(49) **(EG 1)** A meta-analysis of 23 studies (524 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET and PET-CT in Ewing sarcoma family of tumors reported that pooled sensitivities and pooled specificities varied by indication; for identifying recurrence (5 trials), the pooled sensitivity was 93% and the pooled specificity was 90%.(48) **(EG 1)** Expert consensus guidelines and review articles indicate that PET is an option for evaluation of response to chemotherapy.(39)(42)(44)(45) **(EG 2)**

For subsequent evaluation or staging of brain and spinal cord cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A review article and evidence map of clinical studies using PET and PET-CT in glioma found mostly studies assessing diagnostic or prognostic performance, but little evidence demonstrating impact upon diagnostic thinking, therapeutic decisions, or clinical outcomes relevant to patients.(56) **(EG 2)** Malignant glioma, lymphoma, meningioma, or brain metastases may show significantly elevated fluorodeoxyglucose uptake. In addition, fluorodeoxyglucose uptake correlates with tumor grade and prognosis, and may help distinguish radiation necrosis and pseudo-progression from active malignancy.(50)(52) **(EG 2)** A meta-analysis of 26 studies found that PET has moderately good overall accuracy in detecting suspected recurrent glioma following active treatment, with sensitivity of 77% and specificity of 78%.(302) **(EG 1)** A systematic review and meta-analysis of 15 studies (505 patients) evaluating the efficacy of amino acid and F-18 fluorodeoxyglucose PET for differentiating brain metastasis recurrence from radiation necrosis after radiation therapy found a pooled sensitivity and specificity of 85% and 88%, respectively.(303) **(EG 1)**

For subsequent evaluation or staging of breast cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** PET has been shown to be useful for assessing suspected recurrence and metastases, especially when other studies are inconclusive.(59)(306)(307)(308)(309) **(EG 1)** PET-CT has been shown to efficiently detect breast cancer recurrence in asymptomatic patients with rising tumor markers, although its effect upon important outcomes such as survival has yet to be demonstrated.(310) **(EG 2)** Specialty society guidelines therefore indicate that there is no evidence to support the routine use of PET or PET-CT in asymptomatic patients for general post-therapy surveillance.(57)(308)(311) **(EG 2)** Given its high sensitivity and specificity, there may be a role for F-18 fluorodeoxyglucose PET-CT imaging in monitoring response of axillary lymph nodes to neoadjuvant chemotherapy.(312)(313) **(EG 2)** A meta-analysis of 17 studies with a total of 781 patients found that pooled sensitivity of PET-CT for evaluating response to neoadjuvant chemotherapy is 85%, but pooled specificity is only 66%.(314) **(EG 1)** A systematic review and meta-analysis of 26 studies (1752 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET or PET-CT reported a pooled sensitivity and specificity of 90% and 81%, respectively, for the detection of suspected recurrent breast cancer.(315) **(EG 1)** Due to physiologic fluorodeoxyglucose uptake by brain tissue, a principal limitation of PET-CT is lack of sensitivity for metastatic disease to this area.(59) **(EG 2)** F-18 fluorodeoxyglucose PET-CT offers the advantage of being able to assess changes in tumor metabolism after chemotherapy or hormone therapy in patients with metastatic breast cancer.(304) **(EG 2)**

For subsequent evaluation or staging of cervical cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A meta-analysis of 20 studies with a total of 1757 patients found that F-18 fluorodeoxyglucose PET or PET-CT was effective at detecting distant metastasis in recurrent cervical cancer, with pooled sensitivity and specificity of 87% and 97%, respectively; pooled sensitivity and specificity for detecting local regional recurrence were 82% and 98%, respectively.(319) **(EG 1)** A retrospective cohort study of 276 patients with cervical cancer showed that post-treatment PET-CT had overall sensitivity of 95%, specificity of 88%, and accuracy of 90%; the results modified the diagnosis and/or treatment plan in 24% of patients.(318) **(EG 2)** A review article found that sensitivity, specificity, and accuracy of PET-CT for evaluating local recurrence were 82%, 97%, and 92%, respectively, while for evaluation of suspected distant metastases these values were 100%, 90%, and 94%, respectively.(78) **(EG 2)** A systematic review of 15 studies on the use of PET-CT for post-therapy surveillance of asymptomatic patients with cervical cancer found insufficient evidence to draw conclusions regarding efficacy.(320) **(EG 1)**

For subsequent evaluation or staging of colorectal cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** PET-CT is indicated in patients with an unexpected rise in carcinoembryonic antigen after treatment to evaluate for suspected recurrence, if physical examination, colonoscopy, and chest, abdominal, and pelvic CT are negative.(79)(80) **(EG 2)** For detection of liver metastases from colorectal cancer, MRI remains the preferred imaging modality, but given its high sensitivity and specificity, F-18 fluorodeoxyglucose PET can be used as an alternate diagnostic test.(321)(322) **(EG 1)** A systematic review of 12 studies on the use of PET-CT for post-therapy surveillance of asymptomatic patients with colorectal cancer, lymphoma, or head and neck cancer found insufficient evidence to draw conclusions as to efficacy.(323) **(EG 1)** A systematic review and meta-analysis of 10 studies (304 patients, 609 lesions) of various ablation techniques for colorectal cancer liver metastases found that PET-CT had a pooled sensitivity and specificity of 85% and 92%, respectively, for the detection of disease progression after ablation therapy, and that CT scan had a pooled sensitivity and specificity of 53% and 96%, respectively.(324) **(EG 1)** A meta-analysis of 11 studies with a total of 510 patients found that pooled sensitivity of F-18 fluorodeoxyglucose PET or PET-CT for detection of tumor recurrence in colorectal cancer patients with elevated carcinoembryonic antigen was 90.3%; pooled specificity was 80%.(325) **(EG 1)** Expert consensus guidelines do not recommend the routine use of PET or PET-CT for surveillance.(79)(80) **(EG 2)** A specialty society guideline notes that, in patients with colorectal cancer, PET-CT may be appropriate for restaging after local recurrence or to detect metastases, including in cases when tumor markers are increasing and first-line imaging (eg, CT with contrast or MRI) is negative.(81)(296) **(EG 2)**

For subsequent evaluation or staging of endometrial cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A review article concluded that PET-CT is useful in the post-therapy follow-up of patients with endometrial cancer and is reliable for early detection of relapse, although additional studies are needed to confirm the impact of PET-CT on overall survival.(234) **(EG 2)** A systematic review and meta-analysis of 8 studies (378 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET for the detection of recurrent endometrial cancer reported a specificity and sensitivity of 95% and 91%, respectively.(326) **(EG 1)** An expert consensus guideline supports the use of PET-CT as an option for evaluation of suspected recurrence or findings suggestive of metastases.(89) **(EG 2)**

For subsequent evaluation or staging of esophageal or gastroesophageal junction cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** PET or PET-CT is considered by expert consensus guidelines and systematic reviews to be an option for post-treatment assessment.(93)(95)(99) **(EG 1)** A meta-analysis of 8 studies including 486 patients previously treated for esophageal cancer found that F-18 fluorodeoxyglucose PET or PET-CT had pooled sensitivity and specificity of 96% and 78%, respectively, in detecting recurrent disease.(327) **(EG 1)**

For subsequent evaluation or staging of gastric cancer, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** PET or PET-CT is considered by expert consensus guidelines and systematic reviews to be an option for post-treatment assessment.(109)(328)(330) **(EG 1)** A systematic review and meta-analysis of 14 studies (828 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET or PET-CT reported a pooled sensitivity and specificity of 85% and 78%, respectively, for the detection of gastric cancer recurrence.(331) **(EG 1)** A retrospective review of 92 patients suspected of recurrent gastric cancer found sensitivity, specificity, and diagnostic accuracy of PET to be as high as 81%, 87%, and 83%, respectively, especially when recurrent disease was suspected on the basis of other imaging findings or elevated serum tumor markers.(329) **(EG 2)**

For subsequent evaluation or staging of gestational trophoblastic neoplasia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An expert consensus guideline recommends PET-CT for assessment of response to treatment.(332) **(EG 2)**

For subsequent evaluation or staging of head and neck cancer (nonthyroid, non-central nervous system), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Clinical studies and review articles indicate the effectiveness of PET and PET-CT in monitoring therapeutic effectiveness and in surveillance for recurrent disease,(120)(296)(333)(334)(335)(336) including for pediatric patients.(122) **(EG 2)** A systematic review and meta-analysis of 27 studies (1195 patients) evaluating the efficacy of PET or PET-CT for the detection of residual or recurrent head and neck squamous cell carcinoma reported sensitivities and specificities of 86% and 82%, respectively, for residual or recurrent disease at the primary site; 72% and 88%, respectively, for residual or recurrent neck disease; and 85% and 95%, respectively, for the detection of distant metastatic disease.(337) **(EG 1)** A systematic review and meta-analysis of squamous cell carcinoma of the head and neck analyzed 20 studies (1293 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET-CT to assess treatment response and reported a pooled sensitivity and specificity of 85% and 93%, respectively, for detection of residual or recurrent nodal disease; sensitivity and specificity were lower in HPV-positive tumors.(338) **(EG 1)**

For subsequent evaluation or staging of Hodgkin lymphoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** For Hodgkin lymphoma, expert consensus guidelines recommend PET or PET-CT for assessment of response to treatment.(127)(132) **(EG 2)** Review articles and expert opinion indicate that PET is effective in assessment of early response to chemotherapy, radiotherapy planning,(339) post-treatment assessment (including evaluation of residual masses), and follow-up to investigate suspected relapse.(127)(134)(137)(138) **(EG 2)** Specialty society guidelines recommend PET-CT for assessment of response after treatment in fluorodeoxyglucose-avid Hodgkin lymphomas.(128) **(EG 2)** Studies have found that the sensitivity and specificity of F-18 fluorodeoxyglucose PET are significantly higher than those of contrast CT scan alone, with resulting improvements in staging and detecting recurrence; however, improved survival has not been documented.(139) **(EG 2)** A literature review found that indiscriminate routine post-treatment surveillance in Hodgkin lymphoma during first remission, except for higher-risk patients with symptoms or abnormal physical examination, is associated with high false-positive rates and has not been shown to improve survival.(340) **(EG 2)** A multicenter phase III noninferiority trial including 823 patients with advanced Hodgkin lymphoma found that, as compared with standard treatment, interim PET monitoring of chemotherapy response decreased exposure to chemotherapy while maintaining similar 5-year progression-free survival.(341) **(EG 1)** A systematic review of 12 studies on the use of PET-CT for post-therapy surveillance of asymptomatic patients with colorectal cancer, lymphoma, or head and neck cancer found insufficient evidence to draw conclusions as to efficacy.(323) **(EG 1)** A systematic review and meta-analysis of 11 studies (139 patients) of F-18 fluorodeoxyglucose PET followed by biopsy of F-18 fluorodeoxyglucose-avid lesions found a pooled end-of-treatment false-positive rate of 23% for Hodgkin lymphoma.(342) **(EG 1)** A systematic review and meta-analysis of patients with Hodgkin lymphoma found that, as compared with a positive PET or PET-CT, a negative PET or PET-CT performed after initial therapy was associated with improved overall survival (9 studies, 1802 patients) and may be associated with improved progression-free survival (14 studies, 2079 patients).(343) **(EG 1)**

For subsequent evaluation or staging of lung cancer (non-small cell type), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Specialty society guidelines and a systematic review indicate that PET-CT is effective in assessing the results of initial treatment and restaging.(152)(154)(344)(345) **(EG 2)** Fluorodeoxyglucose PET-CT may have a role in differentiating atelectasis and radiation fibrosis from recurrent malignancy, as CT scan may not be able to adequately discriminate between them; however, because fluorodeoxyglucose remains active for up to 2 years after radiation therapy, histopathologic confirmation of recurrent disease is needed.(152) **(EG 2)** A specialty society guideline indicates that PET-CT is insufficient for mediastinal nodal restaging.(167) **(EG 2)** A specialty society guideline notes that fluorodeoxyglucose PET is appropriate when recurrence or progression of non-small cell lung cancer is suspected.(346) **(EG 2)**

For subsequent evaluation or staging of lung cancer (small cell type), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A technology assessment found 3 cohort studies of moderate quality that supported the use of PET or PET-CT for restaging patients to determine the appropriateness of further radiation treatment.(71) **(EG 1)** A study of 29 patients found that the presence of a complete metabolic response on PET-CT (ie, clearing of PET-positive lesions) was associated with longer survival time and may help with therapeutic decision making, although further study is required.(348) **(EG 2)** An expert consensus guideline indicates that PET-CT is not recommended for routine surveillance of patients with small cell lung cancer.(174) **(EG 2)** A specialty society guideline notes that fluorodeoxyglucose PET is appropriate when recurrence or progression of small cell lung cancer is suspected.(346) **(EG 2)**

For subsequent evaluation or staging of melanoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Expert consensus guidelines recommend that PET scan is most useful as a whole-body imaging tool for follow-up of stage III or IV disease,(188) but may also be used at physician discretion for follow-up in stage IIB disease.(184) **(EG 2)** For surveillance of distant metastases in patients with melanoma, a meta-analysis identified 13 studies evaluating PET-CT and found that PET-CT had the highest sensitivity and specificity (86% and 91%, respectively) as compared with the performance of ultrasonography, CT scan, or PET alone.(189) **(EG 1)** A systematic review and meta-analysis including 11 studies (1347 patients) evaluating the accuracy of PET or PET-

CT for diagnosing recurrent disease after treatment of malignant melanoma found a pooled sensitivity of 94% and a pooled specificity of 91%.(351) **(EG 1)**

For subsequent evaluation or staging of multiple myeloma, smoldering myeloma, or solitary plasmacytoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Specialty society guidelines indicate that PET-CT is an option during surveillance and follow-up, especially for unexplained signs or symptoms.(190)(195)(352) **(EG 2)** Fluorodeoxyglucose PET-CT may offer an advantage over MRI or CT scan, as it can differentiate inactive lytic lesions from metabolically active lesions after treatment.(353)(354) **(EG 2)** In a study of 107 myeloma patients treated with autologous stem cell transplant, PET-CT negativity post therapy was significantly correlated with a longer time to relapse (27.6 months) as compared with those with a positive PET-CT (18 months).(355) **(EG 2)**

For subsequent evaluation or staging of neuroendocrine cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A meta-analysis of 22 studies with a total of 2105 patients found that PET-CT (using somatostatin receptor radiotracers) had pooled sensitivity of 93% and pooled specificity of 96% for detection of neuroendocrine tumors in the thorax and abdomen.(208) **(EG 1)** A systematic review of 10 studies with a total of 329 patients found F-18 fluorodeoxyglucose PET and PET-CT to be useful in the detection of nodal and metastatic disease in Merkel cell carcinoma.(210) **(EG 1)** Another systematic review and meta-analysis identified 22 studies evaluating the efficacy of 68Ga-DOTATATE, 68Ga-DOTATOC, or 68Ga-DOTANOC PET or PET-CT for staging and restaging of neuroendocrine tumors and found that PET or PET-CT had a sensitivity range of 78% to 100% and a specificity range of 83% to 100% for the detection of primary and/or metastatic lesions.(213) **(EG 1)** Review articles indicate that use of alternative positron-emitting radiotracers during PET, such as 18F-FDOPA for documented carcinoid tumors or 68Ga-DOTANOC for gastroenteropancreatic neuroendocrine tumors, is highly effective in restaging disease.(19)(202)(216)(356) **(EG 2)** Specialty society and expert consensus guidelines indicate that 68Ga-DOTATATE PET is preferred over indium-111 somatostatin receptor scintigraphy for the subsequent evaluation or staging of gastroenteropancreatic neuroendocrine tumors (eg, gastrinoma, carcinoid, glucagonoma, insulinoma, neuroendocrine tumor of unknown primary site, VIPoma).(23)(205) **(EG 2)** A specialty society guideline recommends F-18 fluorodeoxyglucose PET-CT as the preferred imaging study in patients with known metastatic pheochromocytoma or paraganglioma.(204) **(EG 2)**

For subsequent evaluation or staging of non-Hodgkin lymphoma, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** For B-cell and T-cell lymphoma, an expert consensus guideline recommends the use of PET-CT for diffuse large B-cell lymphoma for the assessment of response to treatment and indicates that the use of PET-CT for other histologies needs to be validated.(219) **(EG 2)** For chronic lymphocytic leukemia, an expert consensus guideline states that PET-CT is generally not useful but can assist in directing nodal biopsy if Richter transformation is suspected.(220) **(EG 2)** For primary cutaneous B-cell lymphoma, an expert consensus guideline recommends PET-CT at the end of treatment to assess response. PET-CT may also be repeated if there is clinical suspicion of progressive disease.(218) **(EG 2)** Review articles and expert opinion indicate that PET is effective in assessment of early response to chemotherapy, radiotherapy planning,(339) post-treatment assessment (including evaluation of residual masses), and follow-up to investigate suspected relapse.(134)(137)(138)(217)(218)(219) **(EG 2)** Specialty society guidelines recommend PET-CT for assessment of response after treatment in fluorodeoxyglucose-avid non-Hodgkin lymphomas, including most histologies except chronic lymphocytic leukemia and small lymphocytic lymphoma, Waldenstrom macroglobulinemia, and mycosis fungoides.(218)(220) **(EG 2)** Studies have found that the sensitivity and specificity of F-18 fluorodeoxyglucose PET are significantly higher than those of contrast CT scan alone, with resulting improvements in staging and detecting recurrence; however, improved survival has not been documented.(139) **(EG 2)** A systematic review of 12 studies on use of PET-CT for post-therapy surveillance of asymptomatic patients with colorectal cancer, lymphoma, or head and neck cancer found insufficient evidence to draw conclusions as to efficacy.(323) **(EG 1)** A systematic review and meta-analysis of 11 studies (139 patients) of F-18 fluorodeoxyglucose PET followed by biopsy of F-18 fluorodeoxyglucose-avid lesions found pooled interim and end-of-treatment false-positive rates of 83% and 32%, respectively.(342) **(EG 1)**

For subsequent evaluation or staging of ovarian cancer (including epithelial ovarian cancer, fallopian tube cancer, and primary peritoneal cancer), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Guidelines suggest PET or PET-CT as an appropriate imaging study for evaluation of suspected recurrent cancer, especially in the setting of rising CA-125 levels.(5)(71)(230) **(EG 2)** PET-CT is also an imaging option for follow-up in those with a complete response to therapy(230) or with possible recurrent disease.(230)(234)(357) **(EG 2)** A systematic review and meta-analysis of 64 studies (3722 patients) compared PET and PET-CT for diagnosis of recurrent or metastatic ovarian cancer and found that PET had a sensitivity and specificity of 89% and 90%, respectively, and that PET-CT had a sensitivity and specificity of 92% and 91%, respectively.(358) **(EG 1)** A specialty society guideline does not support the use of PET for assessing response to treatment or surveillance for nonepithelial ovarian cancer.(233) **(EG 2)** A systematic review of 5 studies (544 patients) found that, as compared with abdominal CT, there was insufficient evidence to support the use of F-18 fluorodeoxyglucose PET-CT or MRI to predict macroscopic incomplete debulking in women with stage III or stage IV ovarian cancer.(359) **(EG 1)**

For subsequent evaluation or staging of pancreatic cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society recommends considering PET-CT in high-risk patients (eg, borderline resectable disease, markedly elevated CA 19-9 level, large primary tumors, or large regional lymph nodes) before and after initiation of neoadjuvant therapy to assess response to systemic therapy and for restaging.(239) **(EG 2)**

For subsequent evaluation or staging of pleural mesothelioma (malignant), evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An expert consensus guideline recommends PET-CT as an adjunct to chest CT after induction chemotherapy to evaluate the mediastinum if there is suspicion for advanced disease in clinical stage I to clinical stage IIIA and epithelioid histology, as well as for radiation therapy treatment planning.(253) **(EG 2)**

For subsequent evaluation or staging of prostate cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review and meta-analysis of C-11 choline and F-18 fluorocholine PET and PET-CT in prostate cancer staging and restaging reported a pooled sensitivity of 85% and a pooled specificity of 88%. Most of the studies identified were retrospective, and concerns regarding the reference standards employed were noted; the authors concluded that the evidence for clinical validity and clinical utility remains to be developed.(259) **(EG 1)** Most prostate cancers are well differentiated and grow slowly with low PET positivity, but the minority of patients with aggressive, rapidly growing tumors may demonstrate greater PET positivity.(262) **(EG 2)** A review article states that, although choline PET-CT has a low sensitivity and specificity for differentiating prostate cancer from normal prostate tissue or hyperplasia, it has a high specificity for detecting bone metastases.(263) **(EG 2)** A specialty society guideline based on a systematic review states that C-11 choline, F-18 fluciclovine, or F-18 sodium fluoride PET-CT may be offered to patients with a persistent or rising PSA and negative conventional imaging (ie, CT, bone scan, or MRI) who are candidates for additional therapy.(264) **(EG 2)** Targeted radionuclide therapy involves conjugation of a radionuclide with a carrier such as an antibody that is targeted to a specific tumor type; the carrier binds to the tumor, allowing the radionuclide to deliver radiation directly to the tumor. PET and PET-CT scans can be used before treatment with targeted radionuclide therapy to determine patient candidacy for such therapy.(297)(298) **(EG 2)** A specialty society states that prostate-specific membrane antigen (PSMA) PET imaging may be used in the initial evaluation of bone and soft tissue metastatic disease, to assess PSA recurrence, or to evaluate disease progression due to its increased sensitivity and specificity as compared with conventional cross-sectional imaging; C-11 choline and F-18 fluciclovine may be used to evaluate small-volume recurrent disease in the bone and soft tissue, while F-18 sodium fluoride is specific to bone metastases and its use is limited to when bone scan results are equivocal.(257) **(EG 2)**

For subsequent evaluation or staging of skin cancer (nonmelanoma), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A review of 40,863 PET studies at 1386 centers, of which 1057 PET studies were for nonmelanoma skin cancer, indicated that PET findings changed management in 23% of those undergoing restaging, but incremental effects of such testing and management changes on longer-term outcomes are unknown.(265) **(EG 2)**

For subsequent evaluation or staging of soft tissue sarcoma, including gastrointestinal stromal tumors, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Fluorodeoxyglucose PET may be useful in prediction of disease progression or recurrence as well as response to chemotherapy.(42)(267)(269)(272) **(EG 2)** For gastrointestinal stromal tumors in particular, PET may be helpful in differentiating active disease from necrotic or scar tissue, malignant from benign tissue, and recurrent tumor from benign changes; however, it is not to be considered a substitute for CT.(268) **(EG 2)** PET-CT is also helpful in early assessment of treatment response for gastrointestinal stromal tumors, especially when there may be secondary resistance to receptor tyrosine kinase (RTK) biotherapy.(274)(275) **(EG 2)**

For subsequent evaluation or staging of testicular cancer (seminoma), evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An expert consensus guideline supports the use of PET after chemotherapy for seminoma if there is a residual mass greater than 3 cm and tumor markers are normal. PET is also indicated for postchemotherapy surveillance of stage IIB, IIC, and III seminoma, as clinically indicated. PET is not clinically indicated for the postdiagnostic evaluation of nonseminomas.(361) **(EG 2)**

For subsequent evaluation or staging of thyroid cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** PET is used for follow-up of patients treated for papillary cancer, follicular cancer, oncocytic carcinoma (Hurthle cell carcinoma), medullary carcinoma, or metastatic anaplastic cancer.(279)(365)(366) **(EG 1)** A specialty society indicates that PET-CT can detect recurrent and metastatic thyroid cancer(285); F-18 fluorodeoxyglucose PET-CT is useful in detecting recurrent differentiated thyroid cancer.(367) **(EG 2)** A systematic review and meta-analysis of 20 studies (958 patients) comparing the efficacy of F-18 fluorodeoxyglucose PET and PET-CT for the detection of recurrent differentiated thyroid cancer found sensitivities of 84% and 92%, respectively, and specificities of 84% and 81%, respectively.(363) **(EG 1)** A systematic review and meta-analysis of 34 studies (2639 patients) evaluating the efficacy of F-18 fluorodeoxyglucose PET for the detection of recurrent differentiated thyroid cancer found a pooled sensitivity and specificity of 79% and 79%, respectively.(368) **(EG 1)**

For subsequent evaluation or staging of vulvar cancer, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** An expert society guideline states that PET-CT is usually appropriate in the post-treatment assessment of suspected recurrent vulvar cancer.(295) **(EG 2)** An expert consensus guideline notes that PET-CT may be used to assess treatment response or with suspected recurrence.(294) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For asymptomatic patients who require screening for cancer, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A comprehensive review of the literature revealed evidence that PET or PET-CT was able to detect subclinical cancer, but studies suffered from multiple confounding variables and sources of bias; in addition, there were concerns about the high cost and limited availability of this technology.(14) **(EG 2)** Given the lack of evidence that PET or PET-CT cancer screening incrementally improves survival, studies have not endorsed PET or PET-CT for this indication.(14) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(369); 42 CFR 422.101(370)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(371); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(372); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(373)

Medicare Coverage Determinations: Medicare Coverage Database(374)

References

1. Krause BJ, Schwarzenbock S, Souvatzoglou M. FDG PET and PET/CT. Recent Results in Cancer Research 2013;187:351-369. DOI: 10.1007/978-3-642-10853-2_12. [Context Link 1] View abstract...
2. Xu G, Zhao L, He Z. Performance of whole-body PET/CT for the detection of distant malignancies in various cancers: a systematic review and meta-analysis. Journal of Nuclear Medicine 2012;53(12):1847-1854. DOI: 10.2967/jnumed.112.105049. [Context Link 1] View abstract...
3. ACR-ACNM-SNMMI-SPR Practice Parameter for Performing FDG-PET/CT in Oncology. Resolution 20 [Internet] American College of Radiology. 2021 Accessed at: https://www.acr.org/. [created 2007; accessed 2024 Sep 19] [Context Link 1]
4. Histed SN, Lindenberg ML, Mena E, Turkbey B, Choyke PL, Kurdziel KA. Review of functional/anatomical imaging in oncology. Nuclear Medicine Communications 2012;33(4):349-361. DOI: 10.1097/MNM.0b013e32834ec8a5. [Context Link 1] View abstract...
5. Decision Memo for Positron Emission Tomography (FDG) for Solid Tumors (CAG-00181R). Medicare Coverage Database [Internet] Centers for Medicare and Medicaid Services. 2009 Apr Accessed at: https://www.cms.gov/medicare-coverage-database/. [accessed 2024 Sep 24] [Context Link 1, 2, 3, 4, 5, 6, 7]
6. Podoloff DA, et al. NCCN task force: clinical utility of PET in a variety of tumor types. Journal of the National Comprehensive Cancer Network 2009;7 Suppl 2:S1-S26. [Context Link 1, 2, 3, 4] View abstract...
7. Bollineni VR, Kramer GM, Jansma EP, Liu Y, Oyen WJ. A systematic review on [(18)F]FLT-PET uptake as a measure of treatment response in cancer patients. European Journal of Cancer 2016;55:81-97. DOI: 10.1016/j.ejca.2015.11.018. [Context Link 1] View abstract...
8. Biermann M, Schwarzlmuller T, Fasmer KE, Reitan BC, Johnsen B, Rosendahl K. Is there a role for PET-CT and SPECT-CT in pediatric oncology? Acta Radiologica 2013;54(9):1037-1045. DOI: 10.1258/ar.2012.120616. [Context Link 1, 2] View abstract...
9. Chambers G, Frood R, Patel C, Scarsbrook A. 18F-FDG PET-CT in paediatric oncology: established and emerging applications. British Journal of Radiology 2019;92(1094):20180584. DOI: 10.1259/bjr.20180584. [Context Link 1, 2, 3, 4, 5, 6, 7] View abstract...
10. Hillner BE, et al. Impact of positron emission tomography/computed tomography and positron emission tomography (PET) alone on expected management of patients with cancer: initial results from the National Oncologic PET Registry. Journal of Clinical Oncology 2008;26(13):2155-2161. DOI: 10.1200/JCO.2007.14.5631. [Context Link 1, 2] View abstract...
11. Hillner BE, et al. The impact of positron emission tomography (PET) on expected management during cancer treatment: findings of the National Oncologic PET Registry. Cancer 2009;115(2):410-418. DOI: 10.1002/cncr.24000. [Context Link 1] View abstract...
12. Huang B, Law MW, Khong PL. Whole-body PET/CT scanning: estimation of radiation dose and cancer risk. Radiology 2009;251(1):166-174. DOI: 10.1148/radiol.2511081300. [Context Link 1] View abstract...
13. Portwine C, Marriott C, Barr RD. PET imaging for pediatric oncology: an assessment of the evidence. Pediatric Blood and Cancer 2010;55(6):1048-1061. DOI: 10.1002/pbc.22747. [Context Link 1] View abstract...
14. Schoder H, Gonen M. Screening for cancer with PET and PET/CT: potential and limitations. Journal of Nuclear Medicine 2007;48 Suppl 1:4S-18S. [Context Link 1, 2] View abstract...
15. Chernyak V, et al. Liver Lesion - Initial Characterization. ACR appropriateness criteria [Internet] American College of Radiology (ACR). 2020 Accessed at: https://www.acr.org. [created 1998; accessed 2024 Sep 26] [Context Link 1]

16. Le Rhun E, et al. EANO-ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up of patients with brain metastasis from solid tumours. Annals of Oncology 2021;32(11):1332-1347. DOI: 10.1016/j.annonc.2021.07.016. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
17. Kim SJ, Lee SW, Pak K, Kim IJ, Kim K. Diagnostic accuracy of 18F-FDG PET or PET/CT for the characterization of adrenal masses: a systematic review and meta-analysis. British Journal of Radiology 2018;91(1086):20170520. DOI: 10.1259/bjr.20170520. [Context Link 1, 2] View abstract...
18. Terzolo M, et al. AME position statement on adrenal incidentaloma. European Journal of Endocrinology 2011;164(6):851-870. DOI: 10.1530/EJE-10-1147. [Context Link 1, 2] View abstract...
19. Wong KK, Arabi M, Zerizer I, Al-Nahhas A, Rubello D, Gross MD. Role of positron emission tomography/computed tomography in adrenal and neuroendocrine tumors: fluorodeoxyglucose and nonfluorodeoxyglucose tracers. Nuclear Medicine Communications 2011;32(9):764-781. DOI: 10.1097/MNM.0b013e3283478a2a. [Context Link 1, 2, 3, 4] View abstract...
20. Taieb D, et al. European Association of Nuclear Medicine Practice Guideline/Society of Nuclear Medicine and Molecular Imaging Procedure Standard 2019 for radionuclide imaging of phaeochromocytoma and paraganglioma. European Journal of Nuclear Medicine and Molecular Imaging 2019;46(10):2112-2137. DOI: 10.1007/s00259-019-04398-1. [Context Link 1] View abstract...
21. Fassnacht M, et al. Adrenocortical carcinomas and malignant phaeochromocytomas: ESMO-EURACAN Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2020;31(11):1476-1490. DOI: 10.1016/j.annonc.2020.08.2099. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
22. Mody RN, et al. Adrenal Mass Evaluation. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2021 Accessed at: https://www.acr.org. [accessed 2024 Sep 19] [Context Link 1]
23. Bergsland E, et al. Neuroendocrine and Adrenal Tumors. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Aug Accessed at: https://www.nccn.org/. [accessed 2024 Aug 16] [Context Link 1, 2, 3, 4, 5, 6, 7]
24. Wong KK, Arabi M, Bou-Assaly W, Marzola MC, Rubello D, Gross MD. Evaluation of incidentally discovered adrenal masses with PET and PET/CT. European Journal of Radiology 2012;81(3):441-450. DOI: 10.1016/j.ejrad.2010.12.060. [Context Link 1] View abstract...
25. Dinnes J, et al. Management of endrocrine disease: Imaging for the diagnosis of malignancy in incidentally discovered adrenal masses: a systematic review and meta-analysis. European Journal of Endocrinology 2016;175(2):R51-R64. DOI: 10.1530/EJE-16-0461. [Context Link 1] View abstract...
26. Benson AB III, et al. Anal Carcinoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2024; 2023 Dec 20 Accessed at: https://www.nccn.org/. [accessed 2024 Sep 19] [Context Link 1, 2, 3, 4]
27. Mahmud A, Poon R, Jonker D. PET imaging in anal canal cancer: a systematic review and meta-analysis. British Journal of Radiology 2017;90(1080):20170370. DOI: 10.1259/bjr.20170370. [Context Link 1, 2, 3, 4, 5] View abstract...
28. Rao S, et al. Anal cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(9):1087-1100. DOI: 10.1016/j.annonc.2021.06.015. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
29. Horvat N, et al. Staging and Follow-up of Anal Cancer. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2024 Accessed at: https://www.acr.org. [created 2024; accessed 2024 Nov 01] [Context Link 1, 2]
30. Flaig TW, et al. Bladder Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 May 09 Accessed at: http://www.nccn.org/. [accessed 2024 Sep 20] [Context Link 1, 2, 3, 4]
31. Powles T, et al. Bladder cancer: ESMO clinical practice guideline for diagnosis, treatment, and follow-up. Annals of Oncology 2022;33(3):244-258. DOI: 10.1016/j.annonc.2021.11.012. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
32. Witjes JA, et al. EAU Guidelines on Muscle-Invasive and Metastatic Bladder Cancer. [Internet] European Association of Urology. 2024 Apr Accessed at: https://uroweb.org/guidelines/. [accessed 2024 Sep 24] [Context Link 1, 2]
33. Comperat E, et al. Current best practice for bladder cancer: a narrative review of diagnostics and treatments. Lancet 2022;400(10364):1712-1721. DOI: 10.1016/S0140-6736(22)01188-6. [Context Link 1] View abstract...
34. Lu YY, et al. Clinical value of FDG PET or PET/CT in urinary bladder cancer: a systemic review and meta-analysis. European Journal of Radiology 2012;81(9):2411-2416. DOI: 10.1016/j.ejrad.2011.07.018. [Context Link 1, 2] View abstract...
35. Ha HK, Koo PJ, Kim SJ. Diagnostic accuracy of F-18 FDG PET/CT for preoperative lymph node staging in newly diagnosed bladder cancer patients: a systematic review and meta-analysis. Oncology 2018;95(1):31-38. DOI: 10.1159/000488200. [Context Link 1] View abstract...
36. Gontero P, et al. EAU Guidelines on Non-Muscle-Invasive Bladder Cancer (TaT1 and CIS). [Internet] European Association of Urology. 2024 Apr Accessed at: https://uroweb.org/. [accessed 2024 Sep 24] [Context Link 1, 2]
37. Chang SS, et al. Treatment of Non-Metastatic Muscle-Invasive Bladder Cancer: AUA/ASCO/ASTRO/SUO Guideline (2017; amended 2020, 2024). Position Statement of the American Urogynecologic Society (AUGS) and the Society for Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU): 2020 Update [Internet] American Urological Association, American Urogynecologic Society. 2024 Apr 25 Accessed at: https://www.auanet.org/. [created 2017; accessed 2024 Oct 02] [Context Link 1, 2]
38. Bladder Cancer: Diagnosis and Management. NICE Guideline NG 2 [Internet] National Institute for Health and Care Excellence. 2015 Feb (NICE Reviewed 2019) Accessed at: https://www.nice.org.uk/guidance/. [accessed 2024 Sep 20] [Context Link 1] View abstract...
39. Biermann JS, et al. Bone Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Mar 12 Accessed at: https://www.nccn.org/. [accessed 2024 Mar 25] [Context Link 1, 2, 3, 4]
40. Strauss SJ, et al. Bone sarcomas: ESMO-EURACAN-GENTURIS-ERNPaedCan Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(12):1520-1536. DOI: 10.1016/j.annonc.2021.08.1995. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
41. Stanborough R, et al. Malignant or Aggressive Primary Musculoskeletal Tumor-Staging And Surveillance. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2022 Accessed at: https://www.acr.org. [created 1998; accessed 2024 Sep 26] [Context Link 1, 2, 3, 4]

42. Garner HW, Kransdorf MJ, Peterson JJ. Posttherapy imaging of musculoskeletal neoplasms. Radiologic Clinics of North America 2011;49(6):1307-1023, vii. DOI: 10.1016/j.rcl.2011.07.011. [Context Link 1, 2, 3, 4] View abstract...

43. Treglia G, Salsano M, Stefanelli A, Mattoli MV, Giordano A, Bonomo L. Diagnostic accuracy of 18F-FDG-PET and PET/CT in patients with Ewing sarcoma family tumours: a systematic review and a meta-analysis. Skeletal Radiology 2012;41(3):249-256. DOI: 10.1007/s00256-011-1298-9. [Context Link 1] View abstract...

44. Costelloe CM, Chuang HH, Madewell JE. FDG PET/CT of primary bone tumors. American Journal of Roentgenology 2014;202(6):W521-W531. DOI: 10.2214/AJR.13.11833. [Context Link 1, 2] View abstract...

45. Cederberg KB, et al. Imaging of pediatric bone tumors: A COG Diagnostic Imaging Committee/SPR Oncology Committee White Paper. Pediatric Blood and Cancer 2023;70 Suppl 4(Suppl 4):Online. DOI: 10.1002/pbc.30000. [Context Link 1, 2] View abstract...

46. Hamel C, et al. Canadian Association of Radiologists Musculoskeletal System Diagnostic Imaging Referral guideline. Canadian Association of Radiologists Journal 2024;75(2):269-278. DOI: 10.1177/08465371231190807. [Context Link 1, 2] View abstract...

47. Ahlawat S, et al. Suspected Primary Bone Tumors. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2024 Accessed at: https://www.acr.org. [created 2024; accessed 2024 Nov 01] [Context Link 1]

48. Huang T, et al. Effectiveness of 18F-FDG PET/CT in the diagnosis, staging and recurrence monitoring of Ewing sarcoma family of tumors: A meta-analysis of 23 studies. Medicine 2018;97(48):e13457. DOI: 10.1097/MD.0000000000013457. [Context Link 1, 2] View abstract...

49. Liu F, Zhang Q, Zhou D, Dong J. Effectiveness of 18F-FDG PET/CT in the diagnosis and staging of osteosarcoma: a meta-analysis of 26 studies. BMC Cancer 2019;19(1):323. DOI: 10.1186/s12885-019-5488-5. [Context Link 1, 2] View abstract...

50. Roelcke U. Imaging brain tumors with PET, SPECT, and ultrasonography. Handbook of Clinical Neurology 2012;104:135-142. DOI: 10.1016/B978-0-444-52138-5.00010-4. [Context Link 1, 2, 3, 4, 5] View abstract...

51. Cornelius JF, Langen KJ, Stoffels G, Hanggi D, Sabel M, Jakob Steiger H. Positron emission tomography imaging of meningioma in clinical practice: review of literature and future directions. Neurosurgery 2012;70(4):1033-1042. DOI: 10.1227/NEU.0b013e31823bcd87. [Context Link 1, 2, 3] View abstract...

52. Nabors LB, et al. Central Nervous System Cancers. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Jul Accessed at: https://www.nccn.org/guidelines/. [accessed 2024 Aug 16] [Context Link 1, 2, 3, 4]

53. Chierichetti F, Pizzolato G. 18F-FDG-PET/CT. Quarterly Journal of Nuclear Medicine and Molecular Imaging 2012;56(2):138-150. [Context Link 1, 2] View abstract...

54. Law I, et al. Joint EANM/EANO/RANO practice guidelines/SNMMI procedure standards for imaging of gliomas using PET with radiolabelled amino acids and [18F]FDG: version 1.0. European Journal of Nuclear Medicine and Molecular Imaging 2019;46(3):540-557. DOI: 10.1007/s00259-018-4207-9. [Context Link 1, 2] View abstract...

55. Treglia G, et al. Diagnostic performance and prognostic value of PET/CT with different tracers for brain tumors: a systematic review of published meta-analyses. International Journal of Molecular Sciences 2019;20(19):4669. DOI: 10.3390/ijms20194669. [Context Link 1, 2] View abstract...

56. Nihashi T, Dahabreh IJ, Terasawa T. PET in the clinical management of glioma: evidence map. American Journal of Roentgenology 2013;200(6):W654-W660. DOI: 10.2214/AJR.12.9168. [Context Link 1, 2] View abstract...

57. Gradishar WJ, et al. Breast Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 Jul 03 Accessed at: https://www.nccn.org/. [accessed 2024 Sep 20] [Context Link 1, 2, 3, 4]

58. Liang X, Yu J, Wen B, Xie J, Cai Q, Yang Q. MRI and FDG-PET/CT based assessment of axillary lymph node metastasis in early breast cancer: a meta-analysis. Clinical Radiology 2017;72(4):295-301. DOI: 10.1016/j.crad.2016.12.001. [Context Link 1, 2] View abstract...

59. Groheux D, Espie M, Giacchetti S, Hindie E. Performance of FDG PET/CT in the clinical management of breast cancer. Radiology 2013;266(2):388-405. DOI: 10.1148/radiol.12110853. [Context Link 1, 2, 3, 4, 5, 6] View abstract...

60. Loibl S, et al. Early breast cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2024;35(2):159-182. DOI: 10.1016/j.annonc.2023.11.016. [Context Link 1] View abstract...

61. Han S, Choi JY. Impact of 18F-FDG PET, PET/CT, and PET/MRI on staging and management as an initial staging modality in breast cancer: a systematic review and meta-analysis. Clinical Nuclear Medicine 2021;46(4):271-282. DOI: 10.1097/RLU.0000000000003502. [Context Link 1] View abstract...

62. Escalona S, Blasco JA, Reza MM, Andradas E, Gomez N. A systematic review of FDG-PET in breast cancer. Medical Oncology 2010;27(1):114-129. DOI: 10.1007/s12032-009-9182-3. [Context Link 1] View abstract...

63. Cooper KL, et al. Positron emission tomography (PET) for assessment of axillary lymph node status in early breast cancer: A systematic review and meta-analysis. European Journal of Surgical Oncology 2011;37(3):187-198. DOI: 10.1016/j.ejso.2011.01.003. [Context Link 1] View abstract...

64. Cooper KL, et al. Positron emission tomography (PET) and magnetic resonance imaging (MRI) for the assessment of axillary lymph node metastases in early breast cancer: systematic review and economic evaluation. Health Technology Assessment 2011;15(4):iii-iv, 1-134. DOI: 10.3310/hta15040. [Context Link 1] View abstract...

65. Sun Z, Yi YL, Liu Y, Xiong JP, He CZ. Comparison of whole-body PET/PET-CT and conventional imaging procedures for distant metastasis staging in patients with breast cancer: a meta-analysis. European Journal of Gynaecological Oncology 2015;36(6):672-676. [Context Link 1] View abstract...

66. Piccardo A, Fiz F, Treglia G, Bottoni G, Trimboli P. Head-to-head comparison between 18F-FES PET/CT and 18F-FDG PET/CT in oestrogen receptor-positive breast Cancer: A Systematic Review and Meta-Analysis. Journal of Clinical Medicine 2022;11(7):Online. DOI: 10.3390/jcm11071919. [Context Link 1] View abstract...

67. Liu T, Cheng T, Xu W, Yan WL, Liu J, Yang HL. A meta-analysis of 18FDG-PET, MRI and bone scintigraphy for diagnosis of bone metastases in patients with breast cancer. Skeletal Radiology 2011;40(5):523-531. DOI: 10.1007/s00256-010-0963-8. [Context Link 1] View abstract...

68. Garcia Vicente AM, et al. Semi-quantitative lymph node assessment of (18)F-FDG PET/CT in locally advanced breast cancer: correlation with biological prognostic factors. European Journal of Nuclear Medicine and Molecular Imaging 2013;40(1):72-79. DOI: 10.1007/s00259-012-2244-3. [Context Link 1] View abstract...
69. Groheux D, et al. 18F-FDG PET/CT in staging patients with locally advanced or inflammatory breast cancer: comparison to conventional staging. Journal of Nuclear Medicine 2013;54(1):5-11. DOI: 10.2967/jnumed.112.106864. [Context Link 1] View abstract...
70. Abu-Rustum NR, et al. Cervical Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 May 06 Accessed at: https://www.nccn.org/. [accessed 2024 May 20] [Context Link 1, 2, 3]
71. Positron Emission Tomography for Nine Cancers (Bladder, Brain, Cervical, Kidney, Ovarian, Pancreatic, Prostate, Small Cell Lung, Testicular). Technology Assessment [Internet] Agency for Healthcare Research and Quality. 2008 Dec Accessed at: https://www.ahrq.gov/. [accessed 2024 Oct 01] [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
72. Mansoori B, Khatri G, Rivera-Colon G, Albuquerque K, Lea J, Pinho DF. Multimodality imaging of uterine cervical malignancies. American Journal of Roentgenology 2020;215(2):292-304. DOI: 10.2214/AJR.19.21941. [Context Link 1, 2, 3] View abstract...
73. Lee SI, Atri M. 2018 FIGO staging system for uterine cervical cancer: enter cross-sectional imaging. Radiology 2019;292(1):15-24. DOI: 10.1148/radiol.2019190088. [Context Link 1] View abstract...
74. Gandy N, Arshad MA, Park WE, Rockall AG, Barwick TD. FDG-PET imaging in cervical cancer. Seminars in Nuclear Medicine 2019;49(6):461-470. DOI: 10.1053/j.semnuclmed.2019.06.007. [Context Link 1] View abstract...
75. Shinagare AB, et al. Pretreatment Evaluation and Follow-up of Invasive Cancer of the Cervix. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2023 Accessed at: https://www.acr.org/. [accessed 2024 Jun 05] [Context Link 1]
76. Gouy S, et al. Nodal-staging surgery for locally advanced cervical cancer in the era of PET. Lancet Oncology 2012;13(5):e212-e220. DOI: 10.1016/S1470-2045(12)70011-6. [Context Link 1] View abstract...
77. Liu B, Gao S, Li S. A comprehensive comparison of CT, MRI, positron emission tomography or positron emission tomography/CT, and diffusion weighted imaging-MRI for detecting the lymph nodes metastases in patients with cervical cancer: a meta-analysis based on 67 studies. Gynecologic and Obstetric Investigation 2017;82(3):209-222. DOI: 10.1159/000456006. [Context Link 1] View abstract...
78. Mirpour S, Mhlanga JC, Logeswaran P, Russo G, Mercier G, Subramaniam RM. The role of PET/CT in the management of cervical cancer. American Journal of Roentgenology 2013;201(2):W192-W205. DOI: 10.2214/AJR.12.9830. [Context Link 1, 2] View abstract...
79. Benson AB III, et al. Colon Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 Jul 03 Accessed at: https://www.nccn.org/. [accessed 2024 Jul 16] [Context Link 1, 2, 3, 4, 5, 6]
80. Benson AB III, et al. Rectal Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Jul 03 Accessed at: https://www.nccn.org/. [accessed 2024 Jul 16] [Context Link 1, 2, 3, 4, 5, 6, 7]
81. Cervantes A, et al. Metastatic colorectal cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2023;34(1):10-32. DOI: 10.1016/j.annonc.2022.10.003. (Reaffirmed 2024 Aug) [Context Link 1, 2, 3, 4] View abstract...
82. You YN, et al. The American Society of Colon and Rectal Surgeons clinical practice guidelines for the management of rectal cancer. Diseases of the Colon and Rectum 2020;63(9):1191-1222. DOI: 10.1097/DCR.0000000000001762. (Reaffirmed 2024 Jul) [Context Link 1, 2] View abstract...
83. Expert Panel on Gastrointestinal Imaging, et al. ACR Appropriateness Criteria® staging of colorectal cancer: 2021 update. Journal of the American College of Radiology 2022;19(5S):S208-S222. DOI: 10.1016/j.jacr.2022.02.012. [Context Link 1] View abstract...
84. Lu YY, Chen JH, Ding HJ, Chien CR, Lin WY, Kao CH. A systematic review and meta-analysis of pretherapeutic lymph node staging of colorectal cancer by 18F-FDG PET or PET/CT. Nuclear Medicine Communications 2012;33(11):1127-1133. DOI: 10.1097/MNM.0b013e328357b2d9. [Context Link 1, 2] View abstract...
85. Chan K, Welch S, Walker-Dilks C, Raifu A, Ontario provincial Gastrointestinal Disease Site Group. Evidence-based guideline recommendations on the use of positron emission tomography imaging in colorectal cancer. Clinical Oncology 2012;24(4):232-249. DOI: 10.1016/j.clon.2011.11.008. [Context Link 1] View abstract...
86. Georgakopoulos A, Pianou N, Kelekis N, Chatziioannou S. Impact of 18F-FDG PET/CT on therapeutic decisions in patients with colorectal cancer and liver metastases. Clinical Imaging 2013;37(3):536-541. DOI: 10.1016/j.clinimag.2012.09.011. [Context Link 1] View abstract...
87. Reinhold C, et al. Pretreatment Evaluation and Follow-Up of Endometrial Cancer. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2020 Accessed at: https://www.acr.org. [created 1999; accessed 2024 Oct 01] [Context Link 1, 2]
88. Oaknin A, et al. Endometrial cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2022;33(9):860-877. DOI: 10.1016/j.annonc.2022.05.009. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
89. Abu-Rustum NR, et al. Uterine Neoplasms. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Mar 06 Accessed at: https://www.nccn.org/. [accessed 2024 Mar 25] [Context Link 1, 2, 3, 4]
90. Hu J, Zhang K, Yan Y, Zang Y, Wang Y, Xue F. Diagnostic accuracy of preoperative 18F-FDG PET or PET/CT in detecting pelvic and para-aortic lymph node metastasis in patients with endometrial cancer: a systematic review and meta-analysis. Archives of Gynecology and Obstetrics 2019;300(3):519-529. DOI: 10.1007/s00404-019-05207-8. [Context Link 1] View abstract...
91. Gee MS, Atri M, Bandos AI, Mannel RS, Gold MA, Lee SI. Identification of distant metastatic disease in uterine cervical and endometrial cancers with FDG PET/CT: analysis from the ACRIN 6671/GOG 0233 multicenter trial. Radiology 2018;287(1):176-184. DOI: 10.1148/radiol.2017170963. [Context Link 1] View abstract...
92. Park JY, et al. The value of preoperative positron emission tomography/computed tomography in node-negative endometrial cancer on magnetic resonance imaging. Annals of Surgical Oncology 2017;24(8):2303-2310. DOI: 10.1245/s10434-017-5901-8. [Context Link 1] View abstract...
93. Ajani JA, et al. Esophageal and Esophagogastric Junction Cancers. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 Jul Accessed at: https://www.nccn.org/. [accessed 2024 Sep 24] [Context Link 1, 2, 3, 4]

94. Oesophago-Gastric Cancer: Assessment and Management in Adults. NICE Guideline NG83 [Internet] National Institute for Health and Care Excellence. 2023 Jul 04 Accessed at: https://www.nice.org.uk/guidance/. [created 2018 Jan 18; accessed 2024 Sep 30] [Context Link 1]
95. Rebollo Aguirre AC, Ramos-Font C, Villegas Portero R, Cook GJ, Llamas Elvira JM, Tabares AR. 18F-fluorodeoxiglucose positron emission tomography for the evaluation of neoadjuvant therapy response in esophageal cancer: systematic review of the literature. Annals of Surgery 2009;250(2):247-254. DOI: 10.1097/SLA.0b013e3181b0c3a6. [Context Link 1, 2, 3] View abstract...
96. Rustgi AK, El-Serag HB. Esophageal carcinoma. New England Journal of Medicine 2014;371(26):2499-2509. DOI: 10.1056/NEJMra1314530. [Context Link 1] View abstract...
97. Obermannova R, et al. Oesophageal cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2022;33(10):992-1004. DOI: 10.1016/j.annonc.2022.07.003. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
98. Raptis CA, et al. Staging and Follow-Up of Esophageal Cancer. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2022 Accessed at: https://www.acr.org/. [created 2022; accessed 2024 Oct 01] [Context Link 1, 2]
99. Marzola MC, et al. Extended staging of oesophageal cancer using FDG-PET - a critical appraisal. European Journal of Radiology 2012;81(1):21-30. DOI: 10.1016/j.ejrad.2010.10.018. [Context Link 1, 2, 3, 4] View abstract...
100. Noble F, Bailey D, SWCIS Upper Gastrointestinal Tumour Panel, Tung K, Byrne JP. Impact of integrated PET/CT in the staging of oesophageal cancer: a UK population-based cohort study. Clinical Radiology 2009;64(7):699-705. DOI: 10.1016/j.crad.2009.03.003. [Context Link 1] View abstract...
101. You JJ, Wong RK, Darling G, Gulenchyn K, Urbain JL, Evans WK. Clinical utility of 18F-fluorodeoxyglucose positron emission tomography/computed tomography in the staging of patients with potentially resectable esophageal cancer. Journal of Thoracic Oncology 2013;8(12):1563-1569. DOI: 10.1097/JTO.0000000000000006. [Context Link 1] View abstract...
102. Moon CM, Bang S, Chung JB. The role of (18)F-fluorodeoxyglucose positron emission tomography in the diagnosis, staging, and follow-up of cholangiocarcinoma. Surgical Oncology 2011;20(1):e10-e17. DOI: 10.1016/j.suronc.2009.08.003. [Context Link 1, 2, 3, 4, 5] View abstract...
103. Sacks A, Peller PJ, Surasi DS, Chatburn L, Mercier G, Subramaniam RM. Value of PET/CT in the management of primary hepatobiliary tumors, part 2. American Journal of Roentgenology 2011;197(2):W260-W265. DOI: 10.2214/AJR.11.6995. [Context Link 1, 2, 3] View abstract...
104. Nagino M, et al. Clinical practice guidelines for the management of biliary tract cancers 2019: The 3rd English edition. Journal of Hepato-Biliary-Pancreatic Sciences 2021;28(1):26-54. DOI: 10.1002/jhbp.870. [Context Link 1] View abstract...
105. Benson AB III, et al. Biliary Tract Cancers. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Jul Accessed at: https://www.nccn.org/. [accessed 2024 Sep 20] [Context Link 1, 2]
106. Vogel A, et al. Biliary tract cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2022;Online. DOI: 10.1016/j.annonc.2022.10.506. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
107. Annunziata S, Pizzuto DA, Caldarella C, Galiandro F, Sadeghi R, Treglia G. Diagnostic accuracy of fluorine-18-fluorodeoxyglucose positron emission tomography in gallbladder cancer: A meta-analysis. World Journal of Gastroenterology 2015;21(40):11481-11488. DOI: 10.3748/wjg.v21.i40.11481. [Context Link 1] View abstract...
108. Annunziata S, et al. Diagnostic accuracy of fluorine-18-fluorodeoxyglucose positron emission tomography in the evaluation of the primary tumor in patients with cholangiocarcinoma: a meta-analysis. BioMed Research International 2014;2014:247693. DOI: 10.1155/2014/247693. [Context Link 1, 2] View abstract...
109. Ajani JA, et al. Gastric Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 Aug Accessed at: https://www.nccn.org/guidelines/. [accessed 2024 Au 16] [Context Link 1, 2, 3, 4]
110. Lordick F, et al. Gastric cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2022;Online. DOI: 10.1016/j.annonc.2022.07.004. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
111. Machiels JP, et al. Squamous cell carcinoma of the oral cavity, larynx, oropharynx and hypopharynx: EHNS-ESMO-ESTRO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2020;31(11):1462-1475. DOI: 10.1016/j.annonc.2020.07.011. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
112. Garg MK, Glanzman J, Kalnicki S. The evolving role of positron emission tomography-computed tomography in organ-preserving treatment of head and neck cancer. Seminars in Nuclear Medicine 2012;42(5):320-327. DOI: 10.1053/j.semnuclmed.2012.04.005. [Context Link 1, 2] View abstract...
113. Xu G, Li J, Zuo X, Li C. Comparison of whole body positron emission tomography (PET)/PET-computed tomography and conventional anatomic imaging for detecting distant malignancies in patients with head and neck cancer: a meta-analysis. Laryngoscope 2012;122(9):1974-1978. DOI: 10.1002/lary.23409. [Context Link 1, 2] View abstract...
114. Pfister DG, et al. Head and Neck Cancers. NCCN Clinical Practice Guideline in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 May 01 Accessed at: https://www.nccn.org/. [accessed 2024 May 20] [Context Link 1, 2, 3]
115. Bossi P, et al. Nasopharyngeal carcinoma: ESMO-EURACAN Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(4):452-465. DOI: 10.1016/j.annonc.2020.12.007. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
116. Hohenstein NA, Chan JW, Wu SY, Tahir P, Yom SS. Diagnosis, staging, radiation treatment response assessment, and outcome prognostication of head and neck cancers using PET imaging: a systematic review. PET Clinics 2020;15(1):65-75. DOI: 10.1016/j.cpet.2019.08.010. [Context Link 1, 2] View abstract...
117. Gule-Monroe MK, et al. Staging and Post-Therapy Assessment of Head and Neck Cancer. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2023 Accessed at: https://www.acr.org/. [accessed 2024 Oct 01] [Context Link 1, 2]
118. Xu GZ, Zhu XD, Li MY. Accuracy of whole-body PET and PET-CT in initial M staging of head and neck cancer: a meta-analysis. Head and Neck 2011;33(1):87-94. DOI: 10.1002/hed.21400. [Context Link 1] View abstract...

119. Xu GZ, Guan DJ, He ZY. (18)FDG-PET/CT for detecting distant metastases and second primary cancers in patients with head and neck cancer. A meta-analysis. Oral Oncology 2011;47(7):560-565. DOI: 10.1016/j.oraloncology.2011.04.021. [Context Link 1] View abstract...

120. Johnson JT, Branstetter BF. PET/CT in head and neck oncology: State-of-the-art 2013. Laryngoscope 2014;124(4):913-915. DOI: 10.1002/lary.23942. [Context Link 1, 2] View abstract...

121. Cheuk DK, et al. PET/CT for staging and follow-up of pediatric nasopharyngeal carcinoma. European Journal of Nuclear Medicine and Molecular Imaging 2012;39(7):1097-1106. DOI: 10.1007/s00259-012-2091-2. [Context Link 1] View abstract...

122. Abramson ZR, et al. Imaging of pediatric head and neck tumors: A COG Diagnostic Imaging Committee/SPR Oncology Committee/ASPNR White Paper. Pediatric Blood and Cancer 2023;70 Suppl 4(Suppl 4):Online. DOI: 10.1002/pbc.30151. [Context Link 1, 2] View abstract...

123. Rohde M, et al. 18F-fluoro-deoxy-glucose-positron emission tomography/computed tomography in diagnosis of head and neck squamous cell carcinoma: a systematic review and meta-analysis. European Journal of Cancer 2014;50(13):2271-2279. DOI: 10.1016/j.ejca.2014.05.015. [Context Link 1] View abstract...

124. Shen G, Zhang W, Jia Z, Li J, Wang Q, Deng H. Meta-analysis of diagnostic value of 18F-FDG PET or PET/CT for detecting lymph node and distant metastases in patients with nasopharyngeal carcinoma. British Journal of Radiology 2014;87(1044):20140296. DOI: 10.1259/bjr.20140296. [Context Link 1] View abstract...

125. Mohandas A, Marcus C, Kang H, Truong MT, Subramaniam RM. FDG PET/CT in the management of nasopharyngeal carcinoma. American Journal of Roentgenology 2014;203(2):W146-W157. DOI: 10.2214/AJR.13.12420. [Context Link 1] View abstract...

126. Abraham J. Imaging for head and neck cancer. Surgical Oncology Clinics of North America 2015;24(3):455-471. DOI: 10.1016/j.soc.2015.03.012. [Context Link 1] View abstract...

127. Hoppe RT, et al. Hodgkin Lymphoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Mar 18 Accessed at: https://www.nccn.org/. [accessed 2024 Mar 25] [Context Link 1, 2, 3, 4, 5]

128. Cheson BD, et al. Recommendations for initial evaluation, staging, and response assessment of Hodgkin and non-Hodgkin lymphoma: the Lugano classification. Journal of Clinical Oncology 2014;32(27):3059-3068. DOI: 10.1200/JCO.2013.54.8800. [Context Link 1, 2, 3, 4, 5, 6, 7] View abstract...

129. Eichenauer DA, et al. Hodgkin lymphoma: ESMO clinical practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2018;29(Suppliment_4):iv19-iv29. DOI: 10.1093/annonc/mdy080. (Reaffirmed 2024 Jul) [Context Link 1, 2] View abstract...

130. Mikhaeel NG, et al. The optimal use of imaging in radiation therapy for lymphoma: guidelines from the International Lymphoma Radiation ncology Group (ILROG). International Journal of Radiation Oncology, Biology and Physics 2019;104(3):501-512. DOI: 10.1016/j.ijrobp.2019.02.001. [Context Link 1, 2, 3] View abstract...

131. Trotman J, Barrington SF. The role of PET in first-line treatment of Hodgkin lymphoma. Lancet. Haematology 2021;8(1):e67-e79. DOI: 10.1016/S2352-3026(20)30357-4. [Context Link 1, 2] View abstract...

132. Mhlanga J, et al. Imaging recommendations in pediatric lymphoma: A COG Diagnostic Imaging Committee/SPR Oncology Committee White Paper. Pediatric Blood and Cancer 2023;70 Suppl 4(Suppl 4):Online. DOI: 10.1002/pbc.29968. [Context Link 1, 2, 3] View abstract...

133. Ilica AT, Kocacelebi K, Savas R, Ayan A. Imaging of extranodal lymphoma with PET/CT. Clinical Nuclear Medicine 2011;36(10):e127-138. DOI: 10.1097/RLU.0b013e31821c99cd. [Context Link 1, 2] View abstract...

134. Ansell SM, Armitage JO. Positron emission tomographic scans in lymphoma: convention and controversy. Mayo Clinic Proceedings 2012;87(6):571-580. DOI: 10.1016/j.mayocp.2012.03.006. [Context Link 1, 2, 3, 4] View abstract...

135. Wu LM, Chen FY, Jiang XX, Gu HY, Yin Y, Xu JR. 18F-FDG PET, combined FDG-PET/CT and MRI for evaluation of bone marrow infiltration in staging of lymphoma: a systematic review and meta-analysis. European Journal of Radiology 2012;81(2):303-311. DOI: 10.1016/j.ejrad.2010.11.020. [Context Link 1, 2] View abstract...

136. Cheng G, Alavi A. Value of 18F-FDG PET versus iliac biopsy in the initial evaluation of bone marrow infiltration in the case of Hodgkin's disease: a meta-analysis. Nuclear Medicine Communications 2013;34(1):25-31. DOI: 10.1097/MNM.0b013e32835afc19. [Context Link 1] View abstract...

137. Qiu L, Chen Y, Wu J. The role of 18F-FDG PET and 18F-FDG PET/CT in the evaluation of pediatric Hodgkin's lymphoma and non-Hodgkin's lymphoma. Hellenic Journal of Nuclear Medicine 2013;16(3):230-236. DOI: 10.1967/s0024499100091. [Context Link 1, 2, 3, 4] View abstract...

138. Johnson SA, Kumar A, Matasar MJ, Schoder H, Rademaker J. Imaging for staging and response assessment in lymphoma. Radiology 2015;276(2):323-338. DOI: 10.1148/radiol.2015142088. [Context Link 1, 2, 3, 4] View abstract...

139. Kostakoglu L, Cheson BD. Current role of FDG PET/CT in lymphoma. European Journal of Nuclear Medicine and Molecular Imaging 2014;41(5):1004-1027. DOI: 10.1007/s00259-013-2686-2. [Context Link 1, 2, 3, 4] View abstract...

140. Rioja J, et al. Role of positron emission tomography in urological oncology. BJU International 2010;106(11):1578-1593. DOI: 10.1111/j.1464-410X.2010.09510.x. [Context Link 1, 2] View abstract...

141. Ozulker T, Ozulker F, Ozbek E, Ozpacaci T. A prospective diagnostic accuracy study of F-18 fluorodeoxyglucose-positron emission tomography/computed tomography in the evaluation of indeterminate renal masses. Nuclear Medicine Communications 2011;32(4):265-272. DOI: 10.1097/MNM.0b013e3283442e3b. [Context Link 1, 2, 3] View abstract...

142. Sacco E, et al. Imaging of renal cell carcinoma: state of the art and recent advances. Urologia Internationalis 2011;86(2):125-139. DOI: 10.1159/000322724. [Context Link 1, 2] View abstract...

143. Motzer RJ, et al. Kidney Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2025; 2024 Jul 01 Accessed at: https://www.nccn.org/. [accessed 2024 Jul 16] [Context Link 1, 2]

144. Campbell S, et al. Renal Mass and Localized Renal Cancer: Evaluation, Management, and Follow-Up. [Internet] American Urological Association. 2021 Apr Accessed at: https://www.auanet.org/. [accessed 2024 Aug 08] [Context Link 1]

145. Wang HY, et al. Meta-analysis of the diagnostic performance of [18F]FDG-PET and PET/CT in renal cell carcinoma. Cancer Imaging 2012;12:464-474. DOI: 10.1102/1470-7330.2012.0042. [Context Link 1] View abstract...
146. Sacks A, Peller PJ, Surasi DS, Chatburn L, Mercier G, Subramaniam RM. Value of PET/CT in the management of liver metastases, part 1. American Journal of Roentgenology 2011;197(2):W256-W259. DOI: 10.2214/AJR.10.6331. [Context Link 1, 2] View abstract...
147. Sharma B, Martin A, Zerizer I. Positron emission tomography-computed tomography in liver imaging. Seminars in Ultrasound, CT and MRI 2013;34(1):66-80. DOI: 10.1053/j.sult.2012.11.006. [Context Link 1] View abstract...
148. Benson AB III, et al. Hepatocellular Carcinoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Jul 02 Accessed at: https://www.nccn.org/. [accessed 2024 Jul 16] [Context Link 1]
149. Ansari D, Keussen I, Andersson R. Positron emission tomography in malignancies of the liver, pancreas and biliary tract - indications and potential pitfalls. Scandinavian Journal of Gastroenterology 2013;48(3):259-265. DOI: 10.3109/00365521.2012.704936. [Context Link 1, 2, 3] View abstract...
150. Lan BY, Kwee SA, Wong LL. Positron emission tomography in hepatobiliary and pancreatic malignancies: a review. American Journal of Surgery 2012;204(2):232-241. DOI: 10.1016/j.amjsurg.2011.07.025. [Context Link 1, 2, 3] View abstract...
151. Schima W, Koelblinger C, Lesnik G, Ba-Ssalamah A. Liver metastases. JBR-BTR : Organe de la Societe Royale Belge de Radiologie (SRBR) 2011 Jul-Aug;94(4):169-177. [Context Link 1] View abstract...
152. Riely GJ, et al. Non-Small Cell Lung Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 8.2024; 2024 Aug 23 Accessed at: https://www.nccn.org/. [accessed 2024 Aug 23] [Context Link 1, 2, 3, 4, 5]
153. Silvestri GA, et al. Methods for staging non-small cell lung cancer: diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2013;143(5 Suppl):e211S-e250S. DOI: 10.1378/chest.12-2355. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
154. Rakheja R, Ko JP, Friedman K. Lung cancer: positron emission tomography/computed tomography and the new staging system. Seminars in Roentgenology 2013;48(4):308-322. DOI: 10.1053/j.ro.2013.05.001. [Context Link 1, 2, 3, 4] View abstract...
155. Hendriks LE, et al. Non-oncogene addicted metastatic non-small-cell lung cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2023;34(4):358-376. DOI: 10.1016/j.annonc.2022.12.013. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
156. Hendriks LE, et al. Oncogene-addicted metastatic non-small-cell lung cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2023;34(4):339-357. DOI: 10.1016/j.annonc.2022.12.009. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...
157. Lung Cancer: Diagnosis and Management. NICE Guidance NG122 [Internet] National Institute for Health and Care Excellence. 2024 Mar Accessed at: https://www.nice.org.uk/guidance. [created 2019; accessed 2024 Sep 26] [Context Link 1, 2]
158. Allen TL, Kendi AT, Mitiek MO, Maddaus MA. Combined contrast-enhanced computed tomography and 18-fluoro-2-deoxy-D-glucose-positron emission tomography in the diagnosis and staging of non-small cell lung cancer. Seminars in Thoracic and Cardiovascular Surgery 2011;23(1):43-50. DOI: 10.1053/j.semtcvs.2011.05.003. [Context Link 1] View abstract...
159. Ambrosini V, et al. PET/CT imaging in different types of lung cancer: an overview. European Journal of Radiology 2012;81(5):988-1001. DOI: 10.1016/j.ejrad.2011.03.020. [Context Link 1, 2] View abstract...
160. Cao JQ, Rodrigues GB, Louie AV, Zaric GS. Systematic review of the cost-effectiveness of positron-emission tomography in staging of non--small-cell lung cancer and management of solitary pulmonary nodules. Clinical Lung Cancer 2012;13(3):161-170. DOI: 10.1016/j.cllc.2011.09.002. [Context Link 1] View abstract...
161. Schmidt-Hansen M, Baldwin DR, Hasler E, Zamora J, Abraira V, Roque I Figuls M. PET-CT for assessing mediastinal lymph node involvement in patients with suspected resectable non-small cell lung cancer. Cochrane Database of Systematic Reviews 2014, (verified by Cochrane 2016 Nov), Issue 11. Art. No.: CD009519. DOI: 10.1002/14651858.CD009519.pub2. [Context Link 1] View abstract...
162. de Groot PM, Carter BW, Betancourt Cuellar SL, Erasmus JJ. Staging of lung cancer. Clinics in Chest Medicine 2015;36(2):179-196, vii-viii. DOI: 10.1016/j.ccm.2015.02.004. [Context Link 1] View abstract...
163. Zhao L, He ZY, Zhong XN, Cui ML. (18)FDG-PET/CT for detection of mediastinal nodal metastasis in non-small cell lung cancer: a meta-analysis. Surgical Oncology 2012;21(3):230-236. DOI: 10.1016/j.suronc.2011.11.001. [Context Link 1] View abstract...
164. Darling GE, et al. Positron emission tomography-computed tomography compared with invasive mediastinal staging in non-small cell lung cancer: results of mediastinal staging in the early lung positron emission tomography trial. Journal of Thoracic Oncology 2011;6(8):1367-1372. DOI: 10.1097/JTO.0b013e318220c912. [Context Link 1] View abstract...
165. Lv YL, et al. Diagnostic performance of integrated positron emission tomography/computed tomography for mediastinal lymph node staging in non-small cell lung cancer: a bivariate systematic review and meta-analysis. Journal of Thoracic Oncology 2011;6(8):1350-1358. DOI: 10.1097/JTO.0b013e31821d4384. [Context Link 1] View abstract...
166. Broderick SR, Patterson GA. Performance of integrated positron emission tomography/computed tomography for mediastinal nodal staging in non-small cell lung carcinoma. Thoracic Surgery Clinics 2013;23(2):193-198. DOI: 10.1016/j.thorsurg.2013.01.014. [Context Link 1] View abstract...
167. De Leyn P, et al. Revised ESTS guidelines for preoperative mediastinal lymph node staging for non-small-cell lung cancer. European Journal of Cardio-Thoracic Surgery 2014;45(5):787-798. DOI: 10.1093/ejcts/ezu028. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...
168. Maziak DE, et al. Positron emission tomography in staging early lung cancer: a randomized trial. Annals of Internal Medicine 2009;151(4):221-228. [Context Link 1] View abstract...
169. Fischer B, et al. Preoperative staging of lung cancer with combined PET-CT. New England Journal of Medicine 2009;361(1):32-39. DOI: 10.1056/NEJMoa0900043. [Context Link 1] View abstract...
170. Mac Manus MP, Hicks RJ. The role of positron emission tomography/computed tomography in radiation therapy planning for patients with lung cancer. Seminars in Nuclear Medicine 2012;42(5):308-319. DOI: 10.1053/j.semnuclmed.2012.04.003. [Context Link 1] View abstract...

171. Chang MC, et al. Meta-analysis: comparison of F-18 fluorodeoxyglucose-positron emission tomography and bone scintigraphy in the detection of bone metastasis in patients with lung cancer. Academic Radiology 2012;19(3):349-357. DOI: 10.1016/j.acra.2011.10.018. [Context Link 1] View abstract...

172. Qu X, Huang X, Yan W, Wu L, Dai K. A meta-analysis of 18FDG-PET-CT, 18FDG-PET, MRI and bone scintigraphy for diagnosis of bone metastases in patients with lung cancer. European Journal of Radiology 2012;81(5):1007-1015. DOI: 10.1016/j.ejrad.2011.01.126. [Context Link 1] View abstract...

173. Paul NS, Ley S, Metser U. Optimal imaging protocols for lung cancer staging: CT, PET, MR imaging, and the role of imaging. Radiologic Clinics of North America 2012;50(5):935-949. DOI: 10.1016/j.rcl.2012.06.007. [Context Link 1] View abstract...

174. Ganti AKP, et al. Small Cell Lung Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Jun Accessed at: https://www.nccn.org/. [accessed 2024 Jun 13] [Context Link 1, 2, 3, 4, 5]

175. Dingemans AC, et al. Small-cell lung cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(7):839-853. DOI: 10.1016/j.annonc.2021.03.207. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

176. Ruben JD, Ball DL. The efficacy of PET staging for small-cell lung cancer: a systematic review and cost analysis in the Australian setting. Journal of Thoracic Oncology 2012;7(6):1015-1020. DOI: 10.1097/JTO.0b013e31824fe90a. [Context Link 1] View abstract...

177. Martin MD, et al. Incidentally Detected Indeterminate Pulmonary Nodule. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2023 Accessed at: https://www.acr.org. [accessed 2024 Sep 26] [Context Link 1, 2]

178. Gould MK, et al. Evaluation of individuals with pulmonary nodules: when is it lung cancer?: diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. Chest 2013;143(5 Suppl):e93S-e120S. DOI: 10.1378/chest.12-2351. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

179. Ruilong Z, Daohai X, Li G, Xiaohong W, Chunjie W, Lei T. Diagnostic value of 18F-FDG-PET/CT for the evaluation of solitary pulmonary nodules: a systematic review and meta-analysis. Nuclear Medicine Communications 2017;38(1):67-75. DOI: 10.1097/MNM.0000000000000605. [Context Link 1, 2] View abstract...

180. Zhao M, Ma Y, Yang B, Wang Y. A meta-analysis to evaluate the diagnostic value of dual-time-point F-fluorodeoxyglucose positron emission tomography/computed tomography for diagnosis of pulmonary nodules. Journal of Cancer Research and Therapeutics 2016;12(Supplement):C304-C308. DOI: 10.4103/0973-1482.200742. [Context Link 1, 2] View abstract...

181. Callister ME, et al. British Thoracic Society guidelines for the investigation and management of pulmonary nodules. Thorax 2015;70 Suppl 2:ii1-ii54. DOI: 10.1136/thoraxjnl-2015-207168. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

182. Furman AM, Dit Yafawi JZ, Soubani AO. An update on the evaluation and management of small pulmonary nodules. Future Oncology 2013;9(6):855-865. DOI: 10.2217/fon.13.17. [Context Link 1] View abstract...

183. Deppen SA, et al. Accuracy of FDG-PET to diagnose lung cancer in areas with infectious lung disease: A meta-analysis. Journal of the American Medical Association 2014;312(12):1227-1236. DOI: 10.1001/jama.2014.11488. [Context Link 1] View abstract...

184. Swetter SM, et al. Melanoma: Cutaneous. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Apr 03 Accessed at: https://www.nccn.org/. [accessed 2024 Apr 29] [Context Link 1, 2, 3, 4, 5]

185. Jimenez-Requena F, et al. Meta-analysis of the performance of (18)F-FDG PET in cutaneous melanoma. European Journal of Nuclear Medicine and Molecular Imaging 2010;37(2):284-300. DOI: 10.1007/s00259-009-1224-8. [Context Link 1] View abstract...

186. Michielin O, van Akkooi ACJ, Ascierto PA, Dummer R, Keilholz U, ESMO Guidelines Committee. Cutaneous melanoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2019;30(12):1884-1901. DOI: 10.1093/annonc/mdz411. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

187. Rohren EM. PET/computed tomography and patient outcomes in melanoma. PET Clinics 2015;10(2):243-554. DOI: 10.1016/j.cpet.2014.12.006. [Context Link 1, 2] View abstract...

188. Keu KV, Iagaru AH. The clinical use of PET/CT in the evaluation of melanoma. Methods in Molecular Biology 2014;1102:553-580. DOI: 10.1007/978-1-62703-727-3_30. [Context Link 1, 2] View abstract...

189. Xing Y, et al. Contemporary diagnostic imaging modalities for the staging and surveillance of melanoma patients: a meta-analysis. Journal of the National Cancer Institute 2011;103(2):129-142. DOI: 10.1093/jnci/djq455. [Context Link 1, 2] View abstract...

190. Kumar SK, et al. Multiple Myeloma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 Apr 26 Accessed at: https://www.nccn.org/. [accessed 2024 Apr 29] [Context Link 1, 2, 3, 4]

191. Baffour FI, Glazebrook KN, Kumar SK, Broski SM. Role of imaging in multiple myeloma. American Journal of Hematology 2020;95(8):966-977. DOI: 10.1002/ajh.25846. [Context Link 1, 2] View abstract...

192. Dimopoulos MA, et al. Multiple myeloma: EHA-ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(3):309-322. DOI: 10.1016/j.annonc.2020.11.014. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

193. Cavo M, et al. Role of (18)F-FDG PET/CT in the diagnosis and management of multiple myeloma and other plasma cell disorders: a consensus statement by the International Myeloma Working Group. Lancet Oncology 2017;18(4):e206-e217. DOI: 10.1016/S1470-2045(17)30189-4. [Context Link 1] View abstract...

194. Chantry A, et al. Guidelines for the use of imaging in the management of patients with myeloma. British Journal of Haematology 2017;178(3):380-393. DOI: 10.1111/bjh.14827. [Context Link 1] View abstract...

195. Barwick T, Bretsztajn L, Wallitt K, Amiras D, Rockall A, Messiou C. Imaging in myeloma with focus on advanced imaging techniques. British Journal of Radiology 2019;92(1095):20180768. DOI: 10.1259/bjr.20180768. [Context Link 1, 2] View abstract...

196. Lu YY, et al. FDG PET or PET/CT for detecting intramedullary and extramedullary lesions in multiple Myeloma: a systematic review and meta-analysis. Clinical Nuclear Medicine 2012;37(9):833-837. DOI: 10.1097/RLU.0b013e31825b2071. [Context Link 1] View abstract...

197. Xia J, et al. Comparison of diagnosing and staging accuracy of PET (CT) and MIBG on patients with neuroblastoma: Systemic review and meta-analysis. Journal of Huazhong University of Science and Technology. Medical Sciences 2017;37(5):649-660. DOI: 10.1007/s11596-017-1785-x. [Context Link 1, 2] View abstract...

198. Park JR, et al. Revisions to the international neuroblastoma response criteria: a consensus statement from the National Cancer Institute Clinical Trials planning meeting. Journal of Clinical Oncology 2017;35(22):2580-2587. DOI: 10.1200/JCO.2016.72.0177. [Context Link 1] View abstract...

199. Kiratli PO, Tuncel M, Bar-Sever Z. Nuclear medicine in pediatric and adolescent tumors. Seminars in Nuclear Medicine 2016;46(4):308-323. DOI: 10.1053/j.semnuclmed.2016.01.004. [Context Link 1] View abstract...

200. Lai HA, et al. Imaging of pediatric neuroblastoma: A COG Diagnostic Imaging Committee/SPR Oncology Committee White Paper. Pediatric Blood and Cancer 2023;70 Suppl 4(Suppl 4):Online. DOI: 10.1002/pbc.29974. [Context Link 1] View abstract...

201. Baudin E, et al. Lung and thymic carcinoids: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(4):439-451. DOI: 10.1016/j.annonc.2021.01.003. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

202. Naswa N, et al. Gallium-68-DOTA-NOC PET/CT of patients with gastroenteropancreatic neuroendocrine tumors: a prospective single-center study. American Journal of Roentgenology 2011;197(5):1221-1228. DOI: 10.2214/AJR.11.7298. [Context Link 1, 2, 3, 4] View abstract...

203. Pavel M, et al. Gastroenteropancreatic neuroendocrine neoplasms: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2020;31(7):844-860. DOI: 10.1016/j.annonc.2020.03.304. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

204. Lenders JW, et al. Pheochromocytoma and paraganglioma: an Endocrine Society clinical practice guideline. Journal of Clinical Endocrinology and Metabolism 2014;99(6):1915-1942. DOI: 10.1210/jc.2014-1498. (Reaffirmed 2024 Aug) [Context Link 1, 2, 3, 4] View abstract...

205. Hope TA, et al. Appropriate use criteria for somatostatin receptor PET imaging in neuroendocrine tumors. Journal of Nuclear Medicine 2020;Nov:66-74. (Reaffirmed 2024 Sep) [Context Link 1, 2, 3, 4]

206. Pollard J, McNeely P, Menda Y. Nuclear imaging of neuroendocrine tumors. Surgical Oncology Clinics of North America 2020;29(2):209-221. DOI: 10.1016/j.soc.2019.11.007. [Context Link 1] View abstract...

207. Fishbein L, et al. The North American Neuroendocrine Tumor Society Consensus Guidelines for Surveillance and Management of Metastatic and/or Unresectable Pheochromocytoma and Paraganglioma. [Internet] North American Neuroendocrine Tumor Society. 2021 Accessed at: https://nanets.net/. [accessed 2024 Oct 02] DOI: 10.1097/MPA.0000000000001792. [Context Link 1] View abstract...

208. Geijer H, Breimer LH. Somatostatin receptor PET/CT in neuroendocrine tumours: update on systematic review and meta-analysis. European Journal of Nuclear Medicine and Molecular Imaging 2013;40(11):1770-1780. DOI: 10.1007/s00259-013-2482-z. [Context Link 1, 2] View abstract...

209. Barrio M, et al. The impact of SSTR-directed PET/CT on the management of patients with neuroendocrine tumor: A systematic review and meta-analysis. Journal of Nuclear Medicine 2017;58(5):756-761. DOI: 10.2967/jnumed.116.185587. [Context Link 1] View abstract...

210. Treglia G, Kakhki VR, Giovanella L, Sadeghi R. Diagnostic performance of fluorine-18-fluorodeoxyglucose positron emission tomography in patients with Merkel cell carcinoma: a systematic review and meta-analysis. American Journal of Clinical Dermatology 2013;14(6):437-447. DOI: 10.1007/s40257-013-0040-x. [Context Link 1, 2] View abstract...

211. Deppen S, et al. 68Ga-DOTATATE compared to 111In-octreotide and conventional imaging for pulmonary and gastroenteropancreatic neuroendocrine tumors: a systematic review and meta-analysis. Journal of Nuclear Medicine 2016;57(6):872-878. DOI: 10.2967/jnumed.115.165803. [Context Link 1] View abstract...

212. Kan Y, Zhang S, Wang W, Liu J, Yang J, Wang Z. 68Ga-somatostatin receptor analogs and 18F-FDG PET/CT in the localization of metastatic pheochromocytomas and paragangliomas with germline mutations: a meta-analysis. Acta Radiologica 2018;59(12):1466-1474. DOI: 10.1177/0284185118764206. [Context Link 1] View abstract...

213. Singh S, Poon R, Wong R, Metser U. 68Ga PET imaging in patients with neuroendocrine tumors: a systematic review and meta-analysis. Clinical Nuclear Medicine 2018;43(11):802-810. DOI: 10.1097/RLU.0000000000002276. [Context Link 1, 2] View abstract...

214. Luo Y, et al. Glucagon-like peptide-1 receptor PET/CT with 68Ga-NOTA-Exendin-4 for detecting localized insulinoma: a prospective cohort study. Journal of Nuclear Medicine 2016;57(5):715-720. DOI: 10.2967/jnumed.115.167445. [Context Link 1] View abstract...

215. Archier A, et al. Prospective comparison of (68)Ga-DOTATATE and (18)F-FDOPA PET/CT in patients with various pheochromocytomas and paragangliomas with emphasis on sporadic cases. European Journal of Nuclear Medicine and Molecular Imaging 2016;43(7):1248-1257. DOI: 10.1007/s00259-015-3268-2. [Context Link 1] View abstract...

216. Lu SJ, Gnanasegaran G, Buscombe J, Navalkissoor S. Single photon emission computed tomography/computed tomography in the evaluation of neuroendocrine tumours: a review of the literature. Nuclear Medicine Communications 2013;34(2):98-107. DOI: 10.1097/MNM.0b013e32835bd59d. [Context Link 1, 2] View abstract...

217. Horwitz SM, et al. T-Cell Lymphomas. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 May Accessed at: https://www.nccn.org/. [accessed 2024 Jun 13] [Context Link 1, 2, 3, 4]

218. Horwitz SM, et al. Primary Cutaneous Lymphomas. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 May 06 Accessed at: https://www.nccn.org/. [accessed 2024 May 20] [Context Link 1, 2, 3, 4, 5, 6]

219. Zelenetz AD, et al. B-Cell Lymphomas. NCCN Clinical Practice Guideline in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Apr 30 Accessed at: https://www.nccn.org/. [accessed 2024 May 20] [Context Link 1, 2, 3, 4, 5]

220. Wierda WG, et al. Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Mar 26 Accessed at: https://www.nccn.org/. [accessed 2024 Apr 29] [Context Link 1, 2, 3, 4]

221. Tilly H, et al. Diffuse large B-cell lymphoma (DLBCL): ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2015;26 Suppl 5:v116-v125. DOI: 10.1093/annonc/mdv304. (Reaffirmed 2024 Jul) [Context Link 1, 2] View abstract...

222. Dreyling M, et al. Newly diagnosed and relapsed follicular lymphoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(3):298-308. DOI: 10.1016/j.annonc.2020.11.008. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

223. Dreyling M, et al. Newly diagnosed and relapsed mantle cell lymphoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2017;28(suppl_4):iv62-iv71. DOI: 10.1093/annonc/mdx223. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

224. Cwynarski K, et al. The management of primary mediastinal B-cell lymphoma: a British Society for Haematology Good Practice Paper. British Journal of Haematology 2019;185(3):402-409. DOI: 10.1111/bjh.15731. (Reaffirmed 2024 Sep) [Context Link 1, 2] View abstract...

225. Kumar SK, et al. Waldenstrom Macroglobulinemia/Lymphoplasmacytic Lymphoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). 2.2024; 2023 Dec 05 Accessed at: https://www.nccn.org/. [accessed 2024 Jan 14] [Context Link 1]

226. Ferreri AJM, et al. Primary central nervous system lymphomas: EHA-ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. HemaSphere 2024;8(6):Online. DOI: 10.1002/hem3.89. [Context Link 1] View abstract...

227. Chen YK, Yeh CL, Tsui CC, Liang JA, Chen JH, Kao CH. F-18 FDG PET for evaluation of bone marrow involvement in non-Hodgkin lymphoma: a meta-analysis. Clinical Nuclear Medicine 2011;36(7):553-559. DOI: 10.1097/RLU.0b013e318217aeff. [Context Link 1] View abstract...

228. Adams HJ, Nievelstein RA, Kwee TC. Systematic Review on the Additional Value of 18F-Fluoro-2-Deoxy-D-Glucose Positron Emission Tomography in Staging Follicular Lymphoma. Journal of Computer Assisted Tomography 2017;41(1):98-103. DOI: 10.1097/RCT.0000000000000485. [Context Link 1] View abstract...

229. Iwamuro M, et al. Diagnostic role of 18F-fluorodeoxyglucose positron emission tomography for follicular lymphoma with gastrointestinal involvement. World Journal of Gastroenterology 2012;18(44):6427-6436. DOI: 10.3748/wjg.v18.i44.6427. [Context Link 1] View abstract...

230. Armstrong DK, et al. Ovarian Cancer Including Fallopian Tube Cancer and Primary Peritoneal Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Jul 15 Accessed at: https://www.nccn.org/. [accessed 2024 Jul 16] [Context Link 1, 2, 3, 4, 5, 6]

231. Viswanathan C, Bhosale PR, Shah SN, Vikram R. Positron emission tomography-computed tomography imaging for malignancies in women. Radiologic Clinics of North America 2013;51(6):1111-1125. DOI: 10.1016/j.rcl.2013.07.006. [Context Link 1, 2] View abstract...

232. Yuan Y, Gu ZX, Tao XF, Liu SY. Computer tomography, magnetic resonance imaging, and positron emission tomography or positron emission tomography/computer tomography for detection of metastatic lymph nodes in patients with ovarian cancer: a meta-analysis. European Journal of Radiology 2012;81(5):1002-1006. DOI: 10.1016/j.ejrad.2011.01.112. [Context Link 1, 2] View abstract...

233. Ray-Coquard I, et al. Non-epithelial ovarian cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2018;29(Supplement 4):iv1-iv18. DOI: 10.1093/annonc/mdy001. (Reaffirmed 2024 Jul) [Context Link 1, 2, 3, 4] View abstract...

234. Dalla Palma M, et al. PET/CT imaging in gynecologic malignancies: a critical overview of its clinical impact and our retrospective single center analysis. Critical Reviews in Oncology/Hematology 2012;83(1):84-98. DOI: 10.1016/j.critrevonc.2011.10.002. [Context Link 1, 2, 3, 4, 5] View abstract...

235. Kitajima K, Murakami K, Sakamoto S, Kaji Y, Sugimura K. Present and future of FDG-PET/CT in ovarian cancer. Annals of Nuclear Medicine 2011;25(3):155-164. DOI: 10.1007/s12149-010-0449-8. [Context Link 1] View abstract...

236. Hynninen J, et al. FDG PET/CT in staging of advanced epithelial ovarian cancer: frequency of supradiaphragmatic lymph node metastasis challenges the traditional pattern of disease spread. Gynecologic Oncology 2012;126(1):64-68. DOI: 10.1016/j.ygyno.2012.04.023. [Context Link 1] View abstract...

237. Duan H, Baratto L, Iagaru A. The role of PET/CT in the imaging of pancreatic neoplasms. Seminars in Ultrasound, CT and MRI 2019;40(6):500-508. DOI: 10.1053/j.sult.2019.04.006. [Context Link 1, 2] View abstract...

238. Panda A, Garg I, Johnson GB, Truty MJ, Halfdanarson TR, Goenka AH. Molecular radionuclide imaging of pancreatic neoplasms. Lancet. Gastroenterology & Hepatology 2019;4(7):559-570. DOI: 10.1016/S2468-1253(19)30081-0. [Context Link 1, 2] View abstract...

239. Tempero MA, et al. Pancreatic Adenocarcinoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Aug Accessed at: https://www.nccn.org/. [accessed 2024 Aug 16] [Context Link 1, 2, 3]

240. Toft J, et al. Imaging modalities in the diagnosis of pancreatic adenocarcinoma: A systematic review and meta-analysis of sensitivity, specificity and diagnostic accuracy. European Journal of Radiology 2017;92:17-23. DOI: 10.1016/j.ejrad.2017.04.009. [Context Link 1] View abstract...

241. Wang Z, Chen JQ, Liu JL, Qin XG, Huang Y. FDG-PET in diagnosis, staging and prognosis of pancreatic carcinoma: a meta-analysis. World Journal of Gastroenterology 2013;19(29):4808-4817. DOI: 10.3748/wjg.v19.i29.4808. [Context Link 1] View abstract...

242. Dibble EH, Karantanis D, Mercier G, Peller PJ, Kachnic LA, Subramaniam RM. PET/CT of cancer patients: part 1, pancreatic neoplasms. American Journal of Roentgenology 2012;199(5):952-967. DOI: 10.2214/AJR.11.8182. [Context Link 1, 2] View abstract...

243. Tang S, et al. Usefulness of 18F-FDG PET, combined FDG-PET/CT and EUS in diagnosing primary pancreatic carcinoma: a meta-analysis. European Journal of Radiology 2011;78(1):142-150. DOI: 10.1016/j.ejrad.2009.09.026. [Context Link 1] View abstract...

244. Raman SP, Horton KM, Fishman EK. Multimodality imaging of pancreatic cancer-computed tomography, magnetic resonance imaging, and positron emission tomography. Cancer Journal 2012;18(6):511-522. DOI: 10.1097/PPO.0b013e318274a461. [Context Link 1] View abstract...

245. Farma JM, et al. PET/CT fusion scan enhances CT staging in patients with pancreatic neoplasms. Annals of Surgical Oncology 2008;15(9):2465-2471. DOI: 10.1245/s10434-008-9992-0. [Context Link 1] View abstract...

246. Conroy T, et al. Pancreatic cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2023;34(11):987-1002. DOI: 10.1016/j.annonc.2023.08.009. (Reaffirmed 2024 Jul) [Context Link 1] View abstract...

247. Pancreatic Cancer in Adults: Diagnosis and Management. NICE Guideline NG85 [Internet] National Institute for Health and Care Excellence. 2018 Feb Accessed at: https://www.nice.org.uk/guidance/. [accessed 2024 Sep 30] [Context Link 1] View abstract...

248. McKeon A, et al. Positron emission tomography-computed tomography in paraneoplastic neurologic disorders: systematic analysis and review. Archives of Neurology 2010;67(3):322-329. DOI: 10.1001/archneurol.2009.336. [Context Link 1, 2] View abstract...

249. Titulaer MJ, et al. Screening for tumours in paraneoplastic syndromes: report of an EFNS task force. European Journal of Neurology 2011;18(1):19-e3. DOI: 10.1111/j.1468-1331.2010.03220.x. [Context Link 1, 2] View abstract...

250. Garcia Vicente AM, et al. 18F-fluorodeoxyglucose positron emission tomography in the diagnosis of malignancy in patients with paraneoplastic neurological syndrome: a systematic review and meta-analysis. European Journal of Nuclear Medicine and Molecular Imaging 2017;44(9):1575-1587. DOI: 10.1007/s00259-017-3722-4. [Context Link 1] View abstract...

251. Sheikhbahaei S, Marcus CV, Fragomeni RS, Rowe SP, Javadi MS, Solnes LB. Whole-body 18F-FDG PET and 18F-FDG PET/CT in patients with suspected paraneoplastic syndrome: a systematic review and meta-analysis of diagnostic accuracy. Journal of Nuclear Medicine 2017;58(7):1031-1036. DOI: 10.2967/jnumed.116.183905. [Context Link 1] View abstract...

252. Kindler HL, et al. Treatment of malignant pleural mesothelioma: American Society of Clinical Oncology clinical practice guideline. Journal of Clinical Oncology 2018;36(13):1343-1373. DOI: 10.1200/JCO.2017.76.6394. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

253. Ettinger DS, et al. Mesothelioma: Pleural. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2024; 2023 Nov 21 Accessed at: https://www.nccn.org/. [accessed 2024 Jan 14] [Context Link 1, 2, 3, 4]

254. Popat S, et al. Malignant pleural mesothelioma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2022;33(2):129-142. DOI: 10.1016/j.annonc.2021.11.005. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

255. Scherpereel A, et al. ERS/ESTS/EACTS/ESTRO guidelines for the management of malignant pleural mesothelioma. European Respiratory Journal 2020;55(6):1900953. DOI: 10.1183/13993003.00953-2019. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

256. Woolhouse I, et al. British Thoracic Society Guideline for the investigation and management of malignant pleural mesothelioma. Thorax 2018;73(Suppl 1):i1-i30. DOI: 10.1136/thoraxjnl-2017-211321. [Context Link 1] View abstract...

257. Schaeffer EM, et al. Prostate Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 May 17 Accessed at: https://www.nccn.org/. [accessed 2024 May 20] [Context Link 1, 2, 3, 4]

258. Akin O, et al. Pretreatment Detection, Surveillance, and Staging of Prostate Cancer. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2022 Accessed at: https://www.acr.org. [created 2005; accessed 2024 Oct 01] [Context Link 1, 2]

259. Umbehr MH, Muntener M, Hany T, Sulser T, Bachmann LM. The role of 11C-choline and 18F-fluorocholine positron emission tomography (PET) and PET/CT in prostate cancer: a systematic review and meta-analysis. European Urology 2013;64(1):106-117. DOI: 10.1016/j.eururo.2013.04.019. [Context Link 1, 2] View abstract...

260. Jeet V, Parkinson B, Song R, Sharma R, Hoyle M. Histopathologically validated diagnostic accuracy of PSMA-PET/CT in the primary and secondary staging of prostate cancer and the impact of PSMA-PET/CT on clinical management: a systematic review and meta-analysis. Seminars in Nuclear Medicine 2023;53(5):706-718. DOI: 10.1053/j.semnuclmed.2023.02.006. [Context Link 1] View abstract...

261. Hope TA, et al. Diagnostic accuracy of 68Ga-PSMA-11 PET for pelvic nodal metastasis detection prior to radical prostatectomy and pelvic lymph node dissection: a multicenter prospective phase 3 Imaging Trial. JAMA Oncology 2021;7(11):1635-1642. DOI: 10.1001/jamaoncol.2021.3771. [Context Link 1] View abstract...

262. Jadvar H. Imaging evaluation of prostate cancer with 18F-fluorodeoxyglucose PET/CT: utility and limitations. European Journal of Nuclear Medicine and Molecular Imaging 2013;40 Suppl 1:S5-S10. DOI: 10.1007/s00259-013-2361-7. [Context Link 1, 2] View abstract...

263. Mapelli P, Picchio M. Initial prostate cancer diagnosis and disease staging--the role of choline-PET-CT. Nature Reviews. Urology 2015;12(9):510-518. DOI: 10.1038/nrurol.2015.191. [Context Link 1, 2] View abstract...

264. Trabulsi EJ, et al. Optimum imaging strategies for advanced prostate cancer: ASCO guideline. Journal of Clinical Oncology 2020;38(17):1963-1996. DOI: 10.1200/JCO.19.02757. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

265. Hillner BE, et al. Relationship between cancer type and impact of PET and PET/CT on intended management: findings of the national oncologic PET registry. Journal of Nuclear Medicine 2008;49(12):1928-1935. DOI: 10.2967/jnumed.108.056713. [Context Link 1, 2, 3, 4, 5] View abstract...

266. Ramelyte E, et al. Swiss recommendations for cutaneous basal cell carcinoma. Dermatology (Basil, Switzerland) 2022;Online. DOI: 10.1159/000526478. [Context Link 1, 2] View abstract...

267. von Mehren M, et al. Soft Tissue Sarcoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 2.2024; 2024 Jul Accessed at: https://www.nccn.org/. [accessed 2024 Aug 16] [Context Link 1, 2, 3, 4]

268. von Mehren M, et al. Gastrointestinal Stromal Tumors (GISTs). NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2024; 2024 Mar 08 Accessed at: https://www.nccn.org/. [accessed 2024 Mar 25] [Context Link 1, 2, 3, 4]

269. Kalkmann J, et al. Consensus report on the radiological management of patients with gastrointestinal stromal tumours (GIST): recommendations of the German GIST Imaging Working Group. Cancer Imaging 2012;12:126-135. DOI: 10.1102/1470-7330.2012.0013. [Context Link 1, 2, 3, 4] View abstract...

270. Casali PG, et al. Gastrointestinal stromal tumours: ESMO-EURACAN-GENTURIS Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2022;33(1):20-33. DOI: 10.1016/j.annonc.2021.09.005. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

271. Gronchi A, et al. Soft tissue and visceral sarcomas: ESMO-EURACAN-GENTURIS Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2021;32(11):1348-1365. DOI: 10.1016/j.annonc.2021.07.006. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

272. Acord MR, et al. Imaging of pediatric extremity soft tissue tumors: A COG Diagnostic Imaging Committee/SPR Oncology Committee White Paper. Pediatric Blood and Cancer 2023;70 Suppl 4(Suppl 4):Online. DOI: 10.1002/pbc.29944. [Context Link 1, 2] View abstract...

273. Etchebehere EC, et al. Assessing the role of 18F-FDG PET and 18F-FDG PET/CT in the diagnosis of soft tissue musculoskeletal malignancies: a systematic review and meta-analysis. European Journal of Nuclear Medicine and Molecular Imaging 2016;43(5):860-870. DOI: 10.1007/s00259-015-3242-z. [Context Link 1] View abstract...

274. Malle P, Sorschag M, Gallowitsch HJ. FDG PET and FDG PET/CT in patients with gastrointestinal stromal tumours. Wiener Medizinische Wochenschrift 2012;162(19-20):423-429. DOI: 10.1007/s10354-012-0131-y. [Context Link 1, 2] View abstract...

275. Tirumani SH, Jagannathan JP, Hornick JL, Ramaiya NH. Resistance to treatment in gastrointestinal stromal tumours: what radiologists should know. Clinical Radiology 2013;68(8):e429-e437. DOI: 10.1016/j.crad.2013.03.016. [Context Link 1, 2] View abstract...

276. Srinivasan A, Parikh A, Pace E, Schechter A, Tang E, Servaes S. Imaging of pediatric abdominal soft tissue tumors: A COG Diagnostic Imaging Committee/SPR Oncology Committee White Paper. Pediatric Blood and Cancer 2023;70 Suppl 4(Suppl 4):Online. DOI: 10.1002/pbc.30341. [Context Link 1] View abstract...

277. Liu Y. Characterization of thymic lesions with F-18 FDG PET-CT: an emphasis on epithelial tumors. Nuclear Medicine Communications 2011;32(7):554-562. DOI: 10.1097/MNM.0b013e328345b984. [Context Link 1, 2] View abstract...

278. Luzzi L, et al. Role of fluorine-flurodeoxyglucose positron emission tomography/computed tomography in preoperative assessment of anterior mediastinal masses. European Journal of Cardio-Thoracic Surgery 2009;36(3):475-479. DOI: 10.1016/j.ejcts.2009.03.055. [Context Link 1, 2, 3] View abstract...

279. Haddad RI, et al. Thyroid Carcinoma. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 3.2024; 2024 Jun 18 Accessed at: https://www.nccn.org/. [accessed 2024 Jul 16] [Context Link 1, 2, 3, 4]

280. Filetti S, et al. Thyroid cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Annals of Oncology 2019;30(12):1856-1883. DOI: 10.1093/annonc/mdz400. (Reaffirmed 2024 Aug) [Context Link 1, 2] View abstract...

281. Kim K, Shim SR, Lee SW, Kim SJ. Diagnostic values of F-18 FDG PET or PET/CT, CT, and US for preoperative lymph node staging in thyroid cancer: a network meta-analysis. British Journal of Radiology 2021;94(1120):Online. DOI: 10.1259/bjr.20201076. [Context Link 1] View abstract...

282. Ho Shon I, et al. Positron emission tomography in non-medullary thyroid cancer. ANZ Journal of Surgery 2011;81(3):116-124. DOI: 10.1111/j.1445-2197.2010.05538.x. [Context Link 1, 2] View abstract...

283. Treglia G, Muoio B, Giovanella L, Salvatori M. The role of positron emission tomography and positron emission tomography/computed tomography in thyroid tumours: an overview. European Archives of Oto-Rhino-Laryngology 2013;270(6):1783-1787. DOI: 10.1007/s00405-012-2205-2. [Context Link 1] View abstract...

284. Riemann B, et al. Diagnostic value and therapeutic impact of (18)F-FDG-PET/CT in differentiated thyroid cancer. Results of a German multicentre study. Nuklearmedizin. Nuclear Medicine 2013;52(1):1-6. DOI: 10.3413/nukmed-0489-12-03. [Context Link 1] View abstract...

285. Yeh MW, et al. American Thyroid Association statement on preoperative imaging for thyroid cancer surgery. Thyroid 2015;25(1):3-14. DOI: 10.1089/thy.2014.0096. (Reaffirmed 2024 Jul) [Context Link 1, 2] View abstract...

286. Johansen J, et al. Prospective study of 18FDG-PET in the detection and management of patients with lymph node metastases to the neck from an unknown primary tumor. Results from the DAHANCA-13 study. Head and Neck 2008;30(4):471-478. DOI: 10.1002/hed.20734. [Context Link 1, 2] View abstract...

287. Dong MJ, Zhao K, Lin XT, Zhao J, Ruan LX, Liu ZF. Role of fluorodeoxyglucose-PET versus fluorodeoxyglucose-PET/computed tomography in detection of unknown primary tumor: a meta-analysis of the literature. Nuclear Medicine Communications 2008;29(9):791-802. DOI: 10.1097/MNM.0b013e328302cd26. [Context Link 1, 2] View abstract...

288. Kwee TC, Kwee RM. Combined FDG-PET/CT for the detection of unknown primary tumors: systematic review and meta-analysis. European Radiology 2009;19(3):731-744. DOI: 10.1007/s00330-008-1194-4. [Context Link 1, 2, 3] View abstract...

289. Stevenson MM, et al. Occult Primary (Cancer of Unknown Primary [CUP]). NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2025; 2024 Aug Accessed at: https://www.nccn.org/. [accessed 2024 Aug 16] [Context Link 1, 2]

290. Woo S, Becker AS, Do RKG, Schoder H, Hricak H, Alberto Vargas H. Impact of 18F-Fluorodeoxyglucose positron emission tomography on management of cancer of unknown primary: systematic review and meta-analysis. European Journal of Cancer 2021;159:60-77. DOI: 10.1016/j.ejca.2021.09.031. [Context Link 1] View abstract...

291. Kramer A, et al. Cancer of unknown primary: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2023;34(3):228-246. DOI: 10.1016/j.annonc.2022.11.013. (Reaffirmed 2024 Jul) [Context Link 1] View abstract...

292. De Dosso S, et al. Detection rate of unknown primary tumour by using somatostatin receptor PET/CT in patients with metastatic neuroendocrine tumours: a meta-analysis. Endocrine 2019;64(3):456-468. DOI: 10.1007/s12020-019-01934-9. [Context Link 1] View abstract...

293. Varadhachary GR, Raber MN. Cancer of unknown primary site. New England Journal of Medicine 2014;371(8):757-765. DOI: 10.1056/NEJMra1303917. [Context Link 1] View abstract...

294. Abu-Rustum NR, et al. Vulvar Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 4.2024; 2024 May 01 Accessed at: https://www.nccn.org/. [accessed 2024 May 20] [Context Link 1, 2, 3, 4]

295. Lakhman Y, et al. Staging and Follow-up of Vulvar Cancer. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2020 Accessed at: https://www.acr.org. [created 2020; accessed 2024 Oct 01] [Context Link 1, 2, 3, 4]

296. Jadvar H, et al. Appropriate use criteria for FDG PET/CT in restaging and treatment response assessment of malignant disease. [Internet] Society of Nuclear Medicine and Medical Imaging. 2018 Jun Accessed at: https://www.snmmi.org/. [created 2017; accessed 2024 Sep 19] DOI: 10.2967/jnumed.117.197988. [Context Link 1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

297. Morris ZS, Wang AZ, Knox SJ. The radiobiology of radiopharmaceuticals. Seminars in Radiation Oncology 2021;31(1):20-27. DOI: 10.1016/j.semradonc.2020.07.002. [Context Link 1, 2] View abstract...

298. Sun J, et al. Imaging-guided targeted radionuclide tumor therapy: From concept to clinical translation. Advanced Drug Delivery Reviews 2022;190:Online. DOI: 10.1016/j.addr.2022.114538. [Context Link 1, 2] View abstract...

299. Rees MA, et al. Imaging of pediatric adrenal tumors: A COG Diagnostic Imaging Committee/SPR Oncology Committee White Paper. Pediatric Blood and Cancer 2023;70 Suppl 4(Suppl 4):Online. DOI: 10.1002/pbc.29973. [Context Link 1] View abstract...

300. Vercellino L, et al. Impact of FDG PET/CT in the staging and the follow-up of anal carcinoma. International Journal of Colorectal Disease 2011;26(2):201-210. DOI: 10.1007/s00384-010-1080-9. [Context Link 1] View abstract...

301. Ivanidze J, et al. Brain Tumors. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2024 Accessed at: https://www.acr.org. [created 2024; accessed 2024 Nov 01] [Context Link 1]

302. Nihashi T, Dahabreh IJ, Terasawa T. Diagnostic accuracy of PET for recurrent glioma diagnosis: a meta-analysis. AJNR. American Journal of Neuroradiology 2013;34(5):944-950, S1-S11. DOI: 10.3174/ajnr.A3324. [Context Link 1] View abstract...

303. Li H, et al. Diagnostic accuracy of amino acid and FDG-PET in differentiating brain metastasis recurrence from radionecrosis after radiotherapy: a systematic review and meta-analysis. AJNR. American Journal of Neuroradiology 2018;39(2):280-288. DOI: 10.3174/ajnr.A5472. [Context Link 1] View abstract...

304. Gold LS, et al. Imaging Techniques for Treatment Evaluation for Metastatic Breast Cancer. Technical Brief No. 17. AHRQ Publication No. 14-EHC044-EF [Internet] Agency for Healthcare Research and Quality Effective Health Care Program. 2014 Oct Accessed at: https://www.effectivehealthcare.ahrq.gov/. [accessed 2024 Sep 26] [Context Link 1, 2] View abstract...

305. Han S, Choi JY. Prognostic value of 18F-FDG PET and PET/CT for assessment of treatment response to neoadjuvant chemotherapy in breast cancer: a systematic review and meta-analysis. Breast Cancer Research 2020;22(1):Online. DOI: 10.1186/s13058-020-01350-2. [Context Link 1] View abstract...

306. Pan L, Han Y, Sun X, Liu J, Gang H. FDG-PET and other imaging modalities for the evaluation of breast cancer recurrence and metastases: a meta-analysis. Journal of Cancer Research and Clinical Oncology 2010;136(7):1007-1022. DOI: 10.1007/s00432-009-0746-6. [Context Link 1] View abstract...

307. Constantinidou A, Martin A, Sharma B, Johnston SR. Positron emission tomography/computed tomography in the management of recurrent/metastatic breast cancer: a large retrospective study from the Royal Marsden Hospital. Annals of Oncology 2011;22(2):307-314. DOI: 10.1093/annonc/mdq343. [Context Link 1] View abstract...

308. McDonald ES, et al. Imaging of Invasive Breast Cancer. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2023 Accessed at: https://www.acr.org/. [accessed 2024 Sep 26] [Context Link 1, 2]

309. Vaz SC, et al. Joint EANM-SNMMI guideline on the role of 2-[18F]FDG PET/CT in no special type breast cancer : (endorsed by the ACR, ESSO, ESTRO, EUSOBI/ESR, and EUSOMA). European Journal of Nuclear Medicine and Molecular Imaging 2024;51(9):2706-2732. DOI: 10.1007/s00259-024-06696-9. [Context Link 1] View abstract...

310. Champion L, et al. Breast cancer recurrence diagnosis suspected on tumor marker rising: value of whole-body 18FDG-PET/CT imaging and impact on patient management. Cancer 2011;117(8):1621-1629. DOI: 10.1002/cncr.25727. [Context Link 1] View abstract...

311. Khatcheressian JL, et al. Breast cancer follow-up and management after primary treatment: American Society of Clinical Oncology clinical practice guideline update. Journal of Clinical Oncology 2013;31(7):961-965. DOI: 10.1200/JCO.2012.45.9859. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

312. Straver ME, et al. Feasibility of FDG PET/CT to monitor the response of axillary lymph node metastases to neoadjuvant chemotherapy in breast cancer patients. European Journal of Nuclear Medicine and Molecular Imaging 2010;37(6):1069-1076. DOI: 10.1007/s00259-009-1343-2. [Context Link 1] View abstract...

313. Groheux D, et al. Triple-negative breast cancer: early assessment with 18F-FDG PET/CT during neoadjuvant chemotherapy identifies patients who are unlikely to achieve a pathologic complete response and are at a high risk of early relapse. Journal of Nuclear Medicine 2012;53(2):249-254. DOI: 10.2967/jnumed.111.094045. [Context Link 1] View abstract...

314. Cheng X, Li Y, Liu B, Xu Z, Bao L, Wang J. 18F-FDG PET/CT and PET for evaluation of pathological response to neoadjuvant chemotherapy in breast cancer: a meta-analysis. Acta Radiologica 2012;53(6):615-627. DOI: 10.1258/ar.2012.110603. [Context Link 1] View abstract...

315. Xiao Y, Wang L, Jiang X, She W, He L, Hu G. Diagnostic efficacy of 18F-FDG-PET or PET/CT in breast cancer with suspected recurrence: a systematic review and meta-analysis. Nuclear Medicine Communications 2016;37(11):1180-1188. DOI: 10.1097/MNM.0000000000000573. [Context Link 1] View abstract...

316. Patel CN, Nazir SA, Khan Z, Gleeson FV, Bradley KM. 18F-FDG PET/CT of cervical carcinoma. American Journal of Roentgenology 2011;196(5):1225-1233. DOI: 10.2214/AJR.10.5084. [Context Link 1] View abstract...

317. Meads C, et al. Positron emission tomography/computerised tomography imaging in detecting and managing recurrent cervical cancer: systematic review of evidence, elicitation of subjective probabilities and economic modelling. Health Technology Assessment 2013;17(12):1-323. DOI: 10.3310/hta17120. [Context Link 1] View abstract...

318. Chung HH, et al. Predictive role of post-treatment [18F]FDG PET/CT in patients with uterine cervical cancer. European Journal of Radiology 2012;81(8):e817-e822. DOI: 10.1016/j.ejrad.2012.02.015. [Context Link 1, 2] View abstract...

319. Chu Y, et al. Diagnostic value of 18F-FDG-PET or PET-CT in recurrent cervical cancer: a systematic review and meta-analysis. Nuclear Medicine Communications 2014;35(2):144-150. DOI: 10.1097/MNM.0000000000000026. [Context Link 1] View abstract...

320. Meads C, et al. Evaluating PET-CT in the detection and management of recurrent cervical cancer: systematic reviews of diagnostic accuracy and subjective elicitation. BJOG: An International Journal of Obstetrics and Gynaecology 2014;121(4):398-407. DOI: 10.1111/1471-0528.12488. [Context Link 1] View abstract...

321. Niekel MC, Bipat S, Stoker J. Diagnostic imaging of colorectal liver metastases with CT, MR imaging, FDG PET, and/or FDG PET/CT: a meta-analysis of prospective studies including patients who have not previously undergone treatment. Radiology 2010;257(3):674-684. DOI: 10.1148/radiol.10100729. [Context Link 1] View abstract...

322. Strasberg SM, Dehdashti F. Role of FDG-PET staging in selecting the optimum patient for hepatic resection of metastatic colorectal cancer. Journal of Surgical Oncology 2010;102(8):955-959. DOI: 10.1002/jso.21729. [Context Link 1] View abstract...

323. Patel K, Hadar N, Lee J, Siegel BA, Hillner BE, Lau J. The lack of evidence for PET or PET/CT surveillance of patients with treated lymphoma, colorectal cancer, and head and neck cancer: a systematic review. Journal of Nuclear Medicine 2013;54(9):1518-1527. DOI:

10.2967/jnumed.112.119362. [Context Link 1, 2, 3] View abstract...

324. Samim M, et al. The diagnostic performance of 18F-FDG PET/CT, CT and MRI in the treatment evaluation of ablation therapy for colorectal liver metastases: A systematic review and meta-analysis. Surgical Oncology 2017;26(1):37-45. DOI: 10.1016/j.suronc.2016.12.006. [Context Link 1] View abstract...
325. Lu YY, et al. Use of FDG-PET or PET/CT to detect recurrent colorectal cancer in patients with elevated CEA: a systematic review and meta-analysis. International Journal of Colorectal Disease 2013;28(8):1039-1047. DOI: 10.1007/s00384-013-1659-z. [Context Link 1] View abstract...
326. Bollineni VR, Ytre-Hauge S, Bollineni-Balabay O, Salvesen HB, Haldorsen IS. High diagnostic value of 18F-FDG PET/CT in endometrial cancer: systematic review and meta-analysis of the literature. Journal of Nuclear Medicine 2016;57(6):879-885. DOI: 10.2967/jnumed.115.170597. [Context Link 1] View abstract...
327. Goense L, et al. Diagnostic performance of 18F-FDG PET and PET/CT for the detection of recurrent esophageal cancer after treatment with curative intent: a systematic review and meta-analysis. Journal of Nuclear Medicine 2015;56(7):995-1002. DOI: 10.2967/jnumed.115.155580. [Context Link 1] View abstract...
328. Shimada H, Okazumi S, Koyama M, Murakami K. Japanese Gastric Cancer Association Task Force for Research Promotion: clinical utility of 18F-fluoro-2-deoxyglucose positron emission tomography in gastric cancer. A systematic review of the literature. Gastric Cancer 2011;14(1):13-21. DOI: 10.1007/s10120-011-0017-5. [Context Link 1, 2] View abstract...
329. Nakamoto Y, et al. Clinical value of whole-body FDG-PET for recurrent gastric cancer: a multicenter study. Japanese Journal of Clinical Oncology 2009;39(5):297-302. DOI: 10.1093/jjco/hyp010. [Context Link 1, 2] View abstract...
330. Wu LM, Hu JN, Hua J, Gu HY, Zhu J, Xu JR. 18 F-fluorodeoxyglucose positron emission tomography to evaluate recurrent gastric cancer: a systematic review and meta-analysis. Journal of Gastroenterology and Hepatology 2012;27(3):472-480. DOI: 10.1111/j.1440-1746.2011.06919.x. [Context Link 1] View abstract...
331. Li P, et al. Fluorine-18-fluorodeoxyglucose positron emission tomography to evaluate recurrent gastric cancer after surgical resection: a systematic review and meta-analysis. Annals of Nuclear Medicine 2016;30(3):179-187. DOI: 10.1007/s12149-016-1058-y. [Context Link 1] View abstract...
332. Abu-Rustum NR, et al. Gestational Trophoblastic Neoplasia. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2024; 2023 Oct 27 Accessed at: https://www.nccn.org/. [accessed 2024 Jan 13] [Context Link 1, 2]
333. Gupta T, et al. Diagnostic performance of post-treatment FDG PET or FDG PET/CT imaging in head and neck cancer: a systematic review and meta-analysis. European Journal of Nuclear Medicine and Molecular Imaging 2011;38(11):2083-2095. DOI: 10.1007/s00259-011-1893-y. [Context Link 1, 2] View abstract...
334. Dunsky KA, Wehrmann DJ, Osman MM, Thornberry BM, Varvares MA. PET-CT and the detection of the asymptomatic recurrence or second primary lesions in the treated head and neck cancer patient. Laryngoscope 2013;123(9):2161-2164. DOI: 10.1002/lary.23941. [Context Link 1] View abstract...
335. Rangaswamy B, et al. Improvement in the detection of locoregional recurrence in head and neck malignancies: F-18 fluorodeoxyglucose-positron emission tomography/computed tomography compared to high-resolution contrast-enhanced computed tomography and endoscopic examination. Laryngoscope 2013;123(11):2664-2669. DOI: 10.1002/lary.24077. [Context Link 1] View abstract...
336. Mehanna H, et al. PET-CT surveillance versus neck dissection in advanced head and neck cancer. New England Journal of Medicine 2016;374(15):1444-1454. DOI: 10.1056/NEJMoa1514493. [Context Link 1] View abstract...
337. Cheung PK, Chin RY, Eslick GD. Detecting residual/recurrent head neck squamous cell carcinomas using PET or PET/CT: systematic review and meta-analysis. Otolaryngology Head and Neck Surgery 2016;154(3):421-432. DOI: 10.1177/0194599815621742. [Context Link 1] View abstract...
338. Helsen N, Van den Wyngaert T, Carp L, Stroobants S. FDG-PET/CT for treatment response assessment in head and neck squamous cell carcinoma: a systematic review and meta-analysis of diagnostic performance. European Journal of Nuclear Medicine and Molecular Imaging 2018;45(6):1063-1071. DOI: 10.1007/s00259-018-3978-3. [Context Link 1] View abstract...
339. Pommier P, et al. Impact of 18F-fluoro-2-deoxyglucose positron emission tomography on treatment strategy and radiotherapy planning for stage I-II Hodgkin disease: a prospective multicenter study. International Journal of Radiation Oncology, Biology and Physics 2011;79(3):823-828. DOI: 10.1016/j.ijrobp.2009.11.048. [Context Link 1, 2] View abstract...
340. El-Galaly TC, et al. Positron emission tomography/computed tomography surveillance in patients with Hodgkin lymphoma in first remission has a low positive predictive value and high costs. Haematologica 2012;97(6):931-936. DOI: 10.3324/haematol.2011.056010. [Context Link 1] View abstract...
341. Casasnovas RO, et al. PET-adapted treatment for newly diagnosed advanced Hodgkin lymphoma (AHL2011): a randomised, multicentre, non-inferiority, phase 3 study. Lancet Oncology 2019;20(2):202-215. DOI: 10.1016/S1470-2045(18)30784-8. [Context Link 1] View abstract...
342. Adams HJ, Kwee TC. Proportion of false-positive lesions at interim and end-of-treatment FDG-PET in lymphoma as determined by histology: Systematic review and meta-analysis. European Journal of Radiology 2016;85(11):1963-1970. DOI: 10.1016/j.ejrad.2016.08.011. [Context Link 1, 2] View abstract...
343. Aldin A, et al. Interim PET-results for prognosis in adults with Hodgkin lymphoma: a systematic review and meta-analysis of prognostic factor studies. Cochrane Database of Systematic Reviews 2020, Issue 1. Art. No.: CD012643. DOI: 10.1002/14651858.CD012643.pub3. [Context Link 1] View abstract...
344. Nair VS, Krupitskaya Y, Gould MK. Positron emission tomography 18F-fluorodeoxyglucose uptake and prognosis in patients with surgically treated, stage I non-small cell lung cancer: a systematic review. Journal of Thoracic Oncology 2009;4(12):1473-1479. DOI: 10.1097/JTO.0b013e3181bccbc6. [Context Link 1, 2] View abstract...
345. Caulo A, Mirsadraee S, Maggi F, Leccisotti L, van Beek EJ, Bonomo L. Integrated imaging of non-small cell lung cancer recurrence: CT and PET-CT findings, possible pitfalls and risk of recurrence criteria. European Radiology 2012;22(3):588-606. DOI: 10.1007/s00330-011-2299-8. [Context

Link 1] View abstract...

346. Mandan R, et al. Lung Cancer-Surveillance After Therapy. ACR Appropriateness Criteria [Internet] American College of Radiology (ACR). 2024 Accessed at: https://www.acr.org. [created 2024; accessed 2024 Nov 01] [Context Link 1, 2]

347. Quartuccio N, et al. The role of 18F-Fluorodeoxyglucose PET/CT in restaging patients with small cell lung cancer: a systematic review. Nuclear Medicine Communications 2021;42(8):839-845. DOI: 10.1097/MNM.0000000000001407. [Context Link 1] View abstract...

348. Ziai D, et al. Therapy response evaluation with FDG-PET/CT in small cell lung cancer: a prognostic and comparison study of the PERCIST and EORTC criteria. Cancer Imaging 2013;13:73-80. DOI: 10.1102/1470-7330.2013.0008. [Context Link 1] View abstract...

349. Michielin O, et al. ESMO consensus conference recommendations on the management of locoregional melanoma: the ESMO Guidelines Committee. Annals of Oncology 2020;31(11):1449-1461. DOI: 10.1016/j.annonc.2020.07.005. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

350. Keilholz U, et al. ESMO consensus conference recommendations on the management of metastatic melanoma: the ESMO Guidelines Committee. Annals of Oncology 2020;31(11):1435-1448. DOI: 10.1016/j.annonc.2020.07.004. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

351. Lee JW, Nam SB, Kim SJ. Role of 18F-fluorodeoxyglucose positron emission tomography or positron emission tomography/computed tomography for the detection of recurrent disease after treatment of malignant melanoma. Oncology 2019;97(5):286-293. DOI: 10.1159/000501398. [Context Link 1] View abstract...

352. de Waal EG, Slart RH, Vellenga E. Is FDG PET a better imaging tool than somatostatin receptor scintigraphy in patients with relapsing multiple myeloma? Clinical Nuclear Medicine 2012;37(10):939-942. DOI: 10.1097/RLU.0b013e3182638e2f. [Context Link 1] View abstract...

353. Koppula B, Kaptuch J, Hanrahan CJ. Imaging of multiple myeloma: usefulness of MRI and PET/CT. Seminars in Ultrasound, CT and MRI 2013;34(6):566-577. DOI: 10.1053/j.sult.2013.05.006. [Context Link 1] View abstract...

354. Barley K, Chari A. Diagnostic advances in multiple myeloma. Current Hematologic Malignancy Reports 2016;11(2):111-117. DOI: 10.1007/s11899-016-0314-5. [Context Link 1] View abstract...

355. Nanni C, et al. The value of 18F-FDG PET/CT after autologous stem cell transplantation (ASCT) in patients affected by multiple myeloma (MM): experience with 77 patients. Clinical Nuclear Medicine 2013;38(2):e74-e79. DOI: 10.1097/RLU.0b013e318266cee2. [Context Link 1] View abstract...

356. Yakemchuk VN, Jager PL, Chirakal R, Reid R, Major P, Gulenchyn KY. PET/CT using 18F-FDOPA provides improved staging of carcinoid tumor patients in a Canadian setting. Nuclear Medicine Communications 2012;33(3):322-330. DOI: 10.1097/MNM.0b013e32834f2603. [Context Link 1, 2] View abstract...

357. Son H, et al. Role of FDG PET/CT in staging of recurrent ovarian cancer. Radiographics 2011;31(2):569-583. DOI: 10.1148/rg.312105713. [Context Link 1] View abstract...

358. Xu B, Ma J, Jiang G, Wang Y, Ma Q. Diagnostic value of positron emission tomography (PET) and PET/computed tomography in recurrent/metastatic ovarian cancer: A meta-analysis. Journal of Obstetrics and Gynaecology Research 2017;43(2):378-386. DOI: 10.1111/jog.13222. [Context Link 1] View abstract...

359. Roze JF, et al. Positron emission tomography (PET) and magnetic resonance imaging (MRI) for assessing tumour resectability in advanced epithelial ovarian/fallopian tube/primary peritoneal cancer. Cochrane Database of Systematic Reviews 2018, Issue 10. Art. No.: CD012567. DOI: 10.1002/14651858.CD012567.pub2. [Context Link 1] View abstract...

360. Lowrance W, et al. Advanced Prostate Cancer. AUA/SUO Guideline [Internet] American Urogynecologic Society. 2023 Accessed at: https://www.auanet.org/. [accessed 2024 Sep 19] [Context Link 1]

361. Gilligan T, et al. Testicular Cancer. NCCN Clinical Practice Guidelines in Oncology [Internet] National Comprehensive Cancer Network (NCCN). v. 1.2024; 2024 Mar 15 Accessed at: https://www.nccn.org/professionals/physician_gls/f_guidelines.asp. [accessed 2024 Mar 25] [Context Link 1, 2]

362. Oldenburg J, et al. Testicular seminoma and non-seminoma: ESMO-EURACAN Clinical Practice Guideline for diagnosis, treatment and follow-up. Annals of Oncology 2022;33(4):362-375. DOI: 10.1016/j.annonc.2022.01.002. (Reaffirmed 2024 Aug) [Context Link 1] View abstract...

363. Caetano R, et al. Accuracy of positron emission tomography and positron emission tomography-CT in the detection of differentiated thyroid cancer recurrence with negative (131) I whole-body scan results: A meta-analysis. Head and Neck 2016;38(2):316-327. DOI: 10.1002/hed.23881. [Context Link 1, 2] View abstract...

364. Chandekar KR, Satapathy S, Bal C. Positron emission tomography/computed tomography in thyroid cancer: an updated review. PET Clinics 2024;19(2):131-145. DOI: 10.1016/j.cpet.2023.12.001. [Context Link 1] View abstract...

365. Cheng X, Bao L, Xu Z, Li D, Wang J, Li Y. 18F-FDG-PET and 18F-FDG-PET/CT in the detection of recurrent or metastatic medullary thyroid carcinoma: a systematic review and meta-analysis. Journal of Medical Imaging and Radiation Oncology 2012;56(2):136-142. DOI: 10.1111/j.1754-9485.2012.02344.x. [Context Link 1] View abstract...

366. Miller ME, Chen Q, Elashoff D, Abemayor E, St John M. Positron emission tomography and positron emission tomography-CT evaluation for recurrent papillary thyroid carcinoma: meta-analysis and literature review. Head and Neck 2011;33(4):562-565. DOI: 10.1002/hed.21492. [Context Link 1] View abstract...

367. Marcus C, Whitworth PW, Surasi DS, Pai SI, Subramaniam RM. PET/CT in the management of thyroid cancers. American Journal of Roentgenology 2014;202(6):1316-1329. DOI: 10.2214/AJR.13.11673. [Context Link 1] View abstract...

368. Haslerud T, et al. F18-FDG-PET for recurrent differentiated thyroid cancer: a systematic meta-analysis. Acta Radiologica 2016;57(10):1193-1200. DOI: 10.1177/0284185115594645. [Context Link 1] View abstract...

369. Centers for Medicare and Medicaid Services. "Hospital services excluded from payment under the hospital outpatient prospective payment system." 42 CFR 419.22 Washington, DC 2023 Jul [accessed 2024 Oct 08] Accessed at: https://www.ecfr.gov/. [Context Link 1]

370. Centers for Medicare and Medicaid Services. "Requirements relating to basic benefits." 42 CFR 422.101 Washington, DC 2024 Sep 30 [accessed 2024 Oct 08] Accessed at: https://www.ecfr.gov/. [Context Link 1]
371. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 14 - Medical Devices Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/. [accessed 2024 Sep 26] [Context Link 1]
372. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 15 - Covered Medical and Other Health Services Rev. 12425 [Internet] Centers for Medicare & Medicaid Services. 2023 Dec 21 Accessed at: https://www.cms.gov/manuals/. [accessed 2024 Feb 01] [Context Link 1]
373. Centers for Medicare and Medicaid Services. Medicare Benefit Policy Manual. Chapter 16 - General Exclusions From Coverage Rev. 198 [Internet] Centers for Medicare and Medicaid Services. 2014 Nov 06 Accessed at: https://www.cms.gov/Regulations-and-Guidance/Guidance/Manuals/. [accessed 2024 Jun 06] [Context Link 1]
374. Medicare Coverage Database. [Internet] Centers for Medicare and Medicaid Services. Accessed at: https://www.cms.gov/medicare-coverage-database/search.aspx? Updated 2024 [accessed 2024 Sep 26] [Context Link 1]

Codes

CPT®: 78811, 78812, 78813, 78814, 78815, 78816, 79101

HCPCS: A9515, A9552, A9587, A9588, A9593, A9594, A9595, A9596, A9597, A9607, A9608, A9609, A9800, G0219, G0235, G0252 [Hide]

MCG Health
Ambulatory Care 29th Edition

Last Update: 6/13/2025 2:16:04 AM
Build Number: 29.1.20250613001137.022065